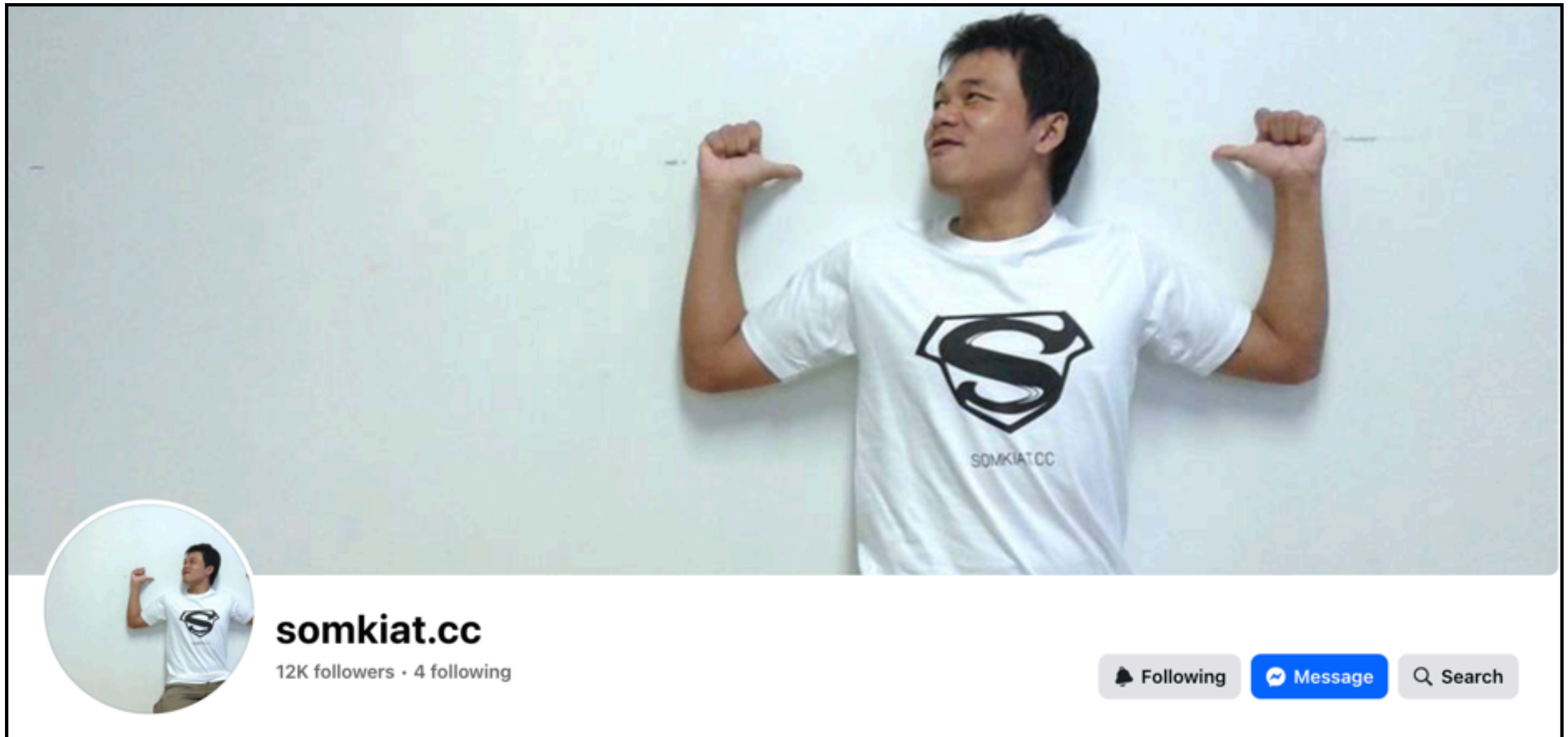




AI Agent for Coding Code Smarter, Not Harder





Software Developer at SCK



**Slide นี้ คนทำนะครับ
ไม่ได้ใช้ Gen AI !!**



AI Agent for Coding

Code smarter, Not harder



Topics

Software development workflows with AI Agent
Spec-Driven Development (SDD)
Demo coding with AI Agent

Requirement

Planning

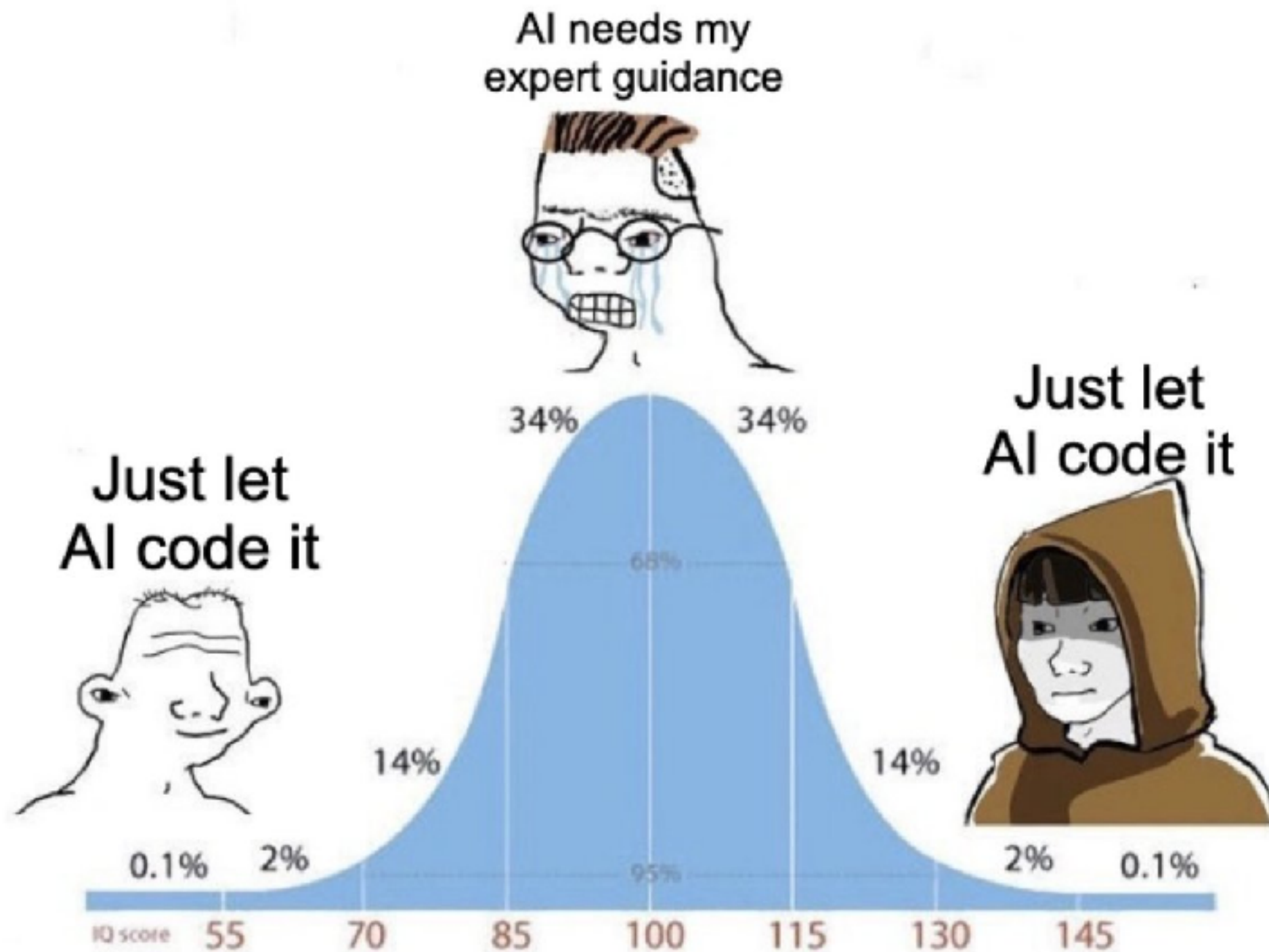
Coding
Testing

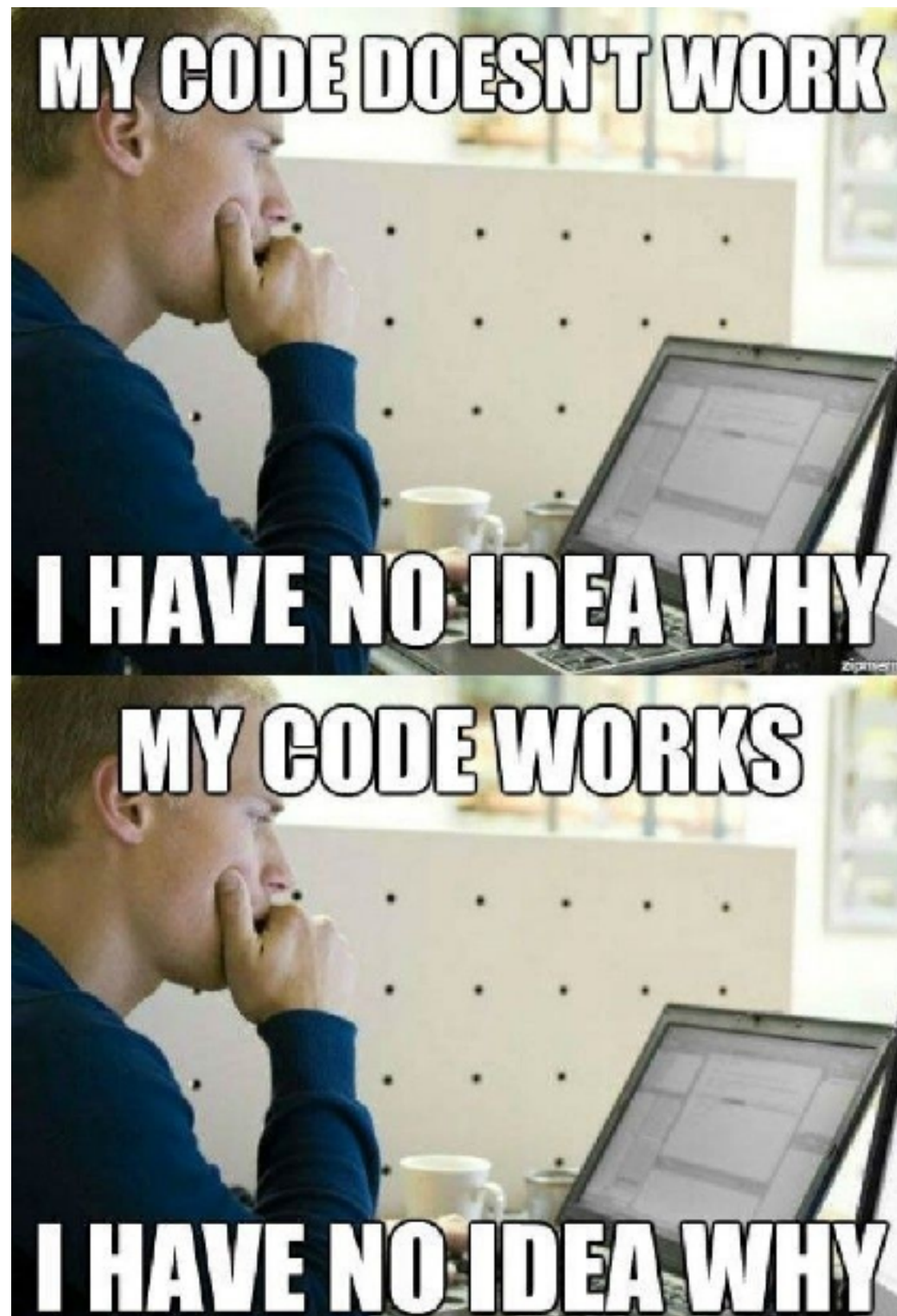
Validate and Testing



Experiment > Optimize







History of Programming



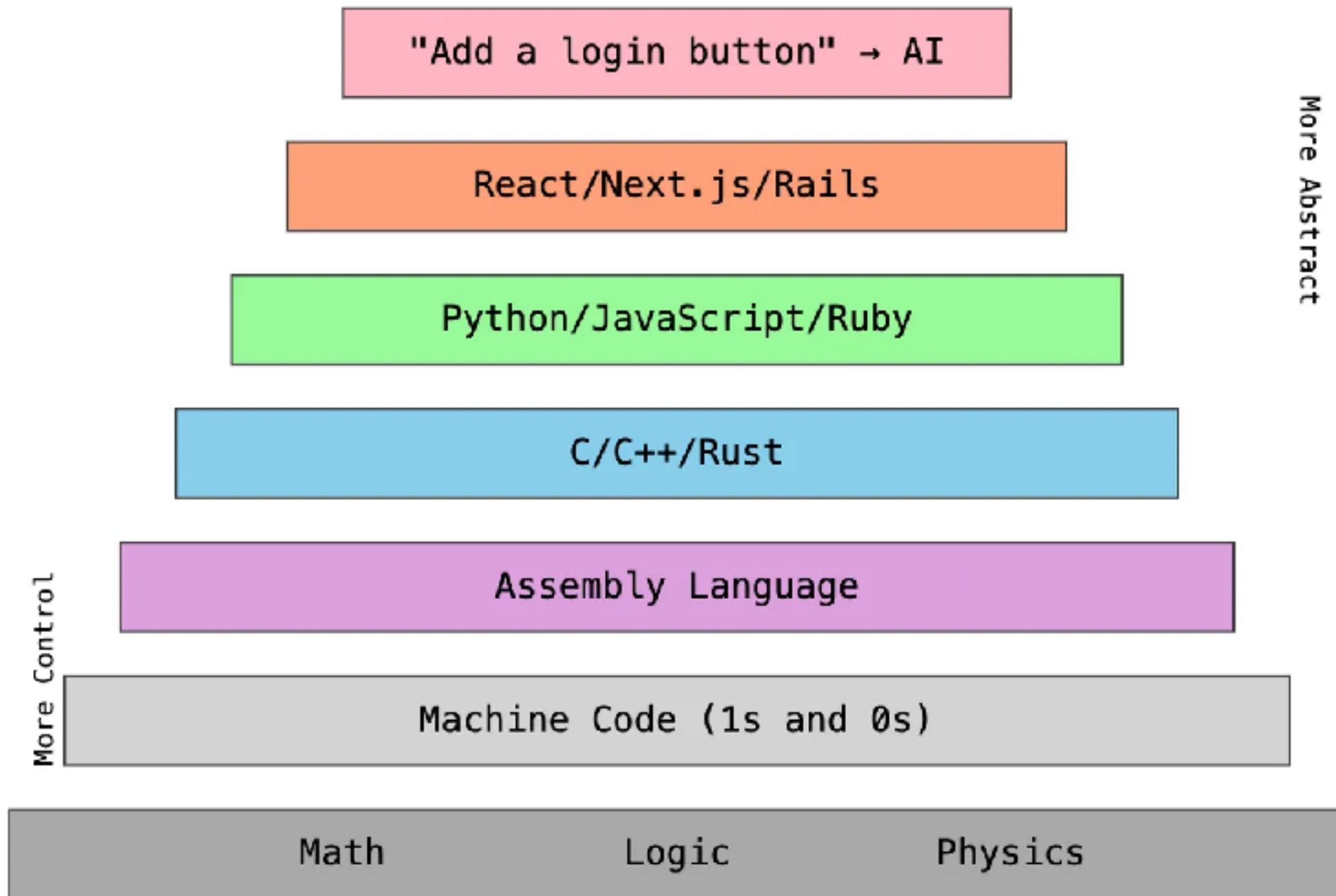
History of Programming



Increase of abstraction
Productivity
Knowledge gap !!



The Modern Tower of Abstraction



<https://cline.bot/blog/from-assembly-to-ai-why-vibe-coding-is-just-another-chapter-in-our-abstraction-story>





Andrej Karpathy ✓

@karpathy



There's a new kind of coding I call "vibe coding", where you fully give in to the vibes, embrace exponentials, and forget that the code even exists. It's possible because the LLMs (e.g. Cursor Composer w Sonnet) are getting too good. Also I just talk to Composer with SuperWhisper so I barely even touch the keyboard. I ask for the dumbest things like "decrease the padding on the sidebar by half" because I'm too lazy to find it. I "Accept All" always, I don't read the diffs anymore. When I get error messages I just copy paste them in with no comment, usually that fixes it. The code grows beyond my usual comprehension, I'd have to really read through it for a while. Sometimes the LLMs can't fix a bug so I just work around it or ask for random changes until it goes away. It's not too bad for throwaway weekend projects, but still quite amusing. I'm building a project or webapp, but it's not really coding - I just see stuff, say stuff, run stuff, and copy paste stuff, and it mostly works.

6:17 AM · Feb 3, 2025 · 5.3M Views

<https://x.com/karpathy/status/1886192184808149383>



Vibe Coding

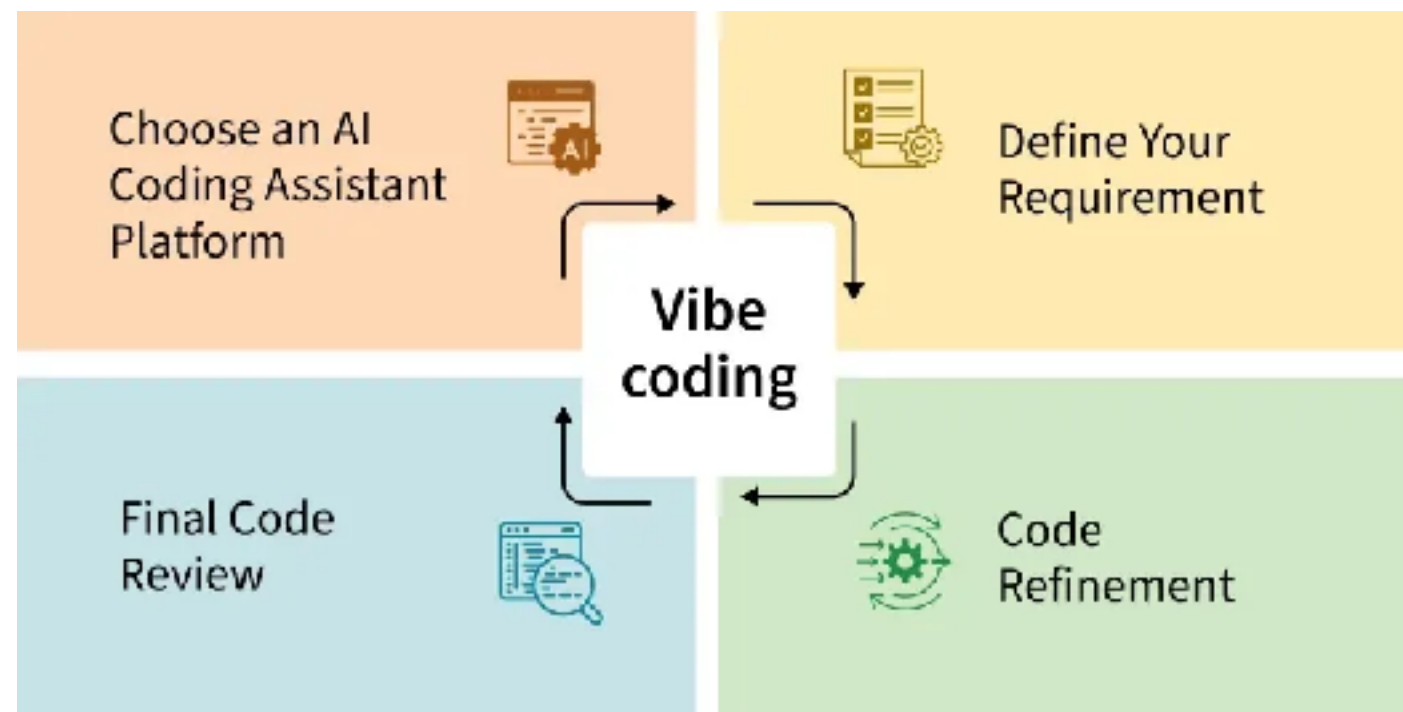
Software development practice that uses artificial intelligence (AI) to generate functional code from **natural language prompts**, accelerating development, and making app building more accessible, especially for those with limited programming experience

<https://cloud.google.com/discover/what-is-vibe-coding>



Types of Vibe Coding ?

Pure vibe coding
Responsible AI-assist development



<https://cloud.google.com/discover/what-is-vibe-coding>



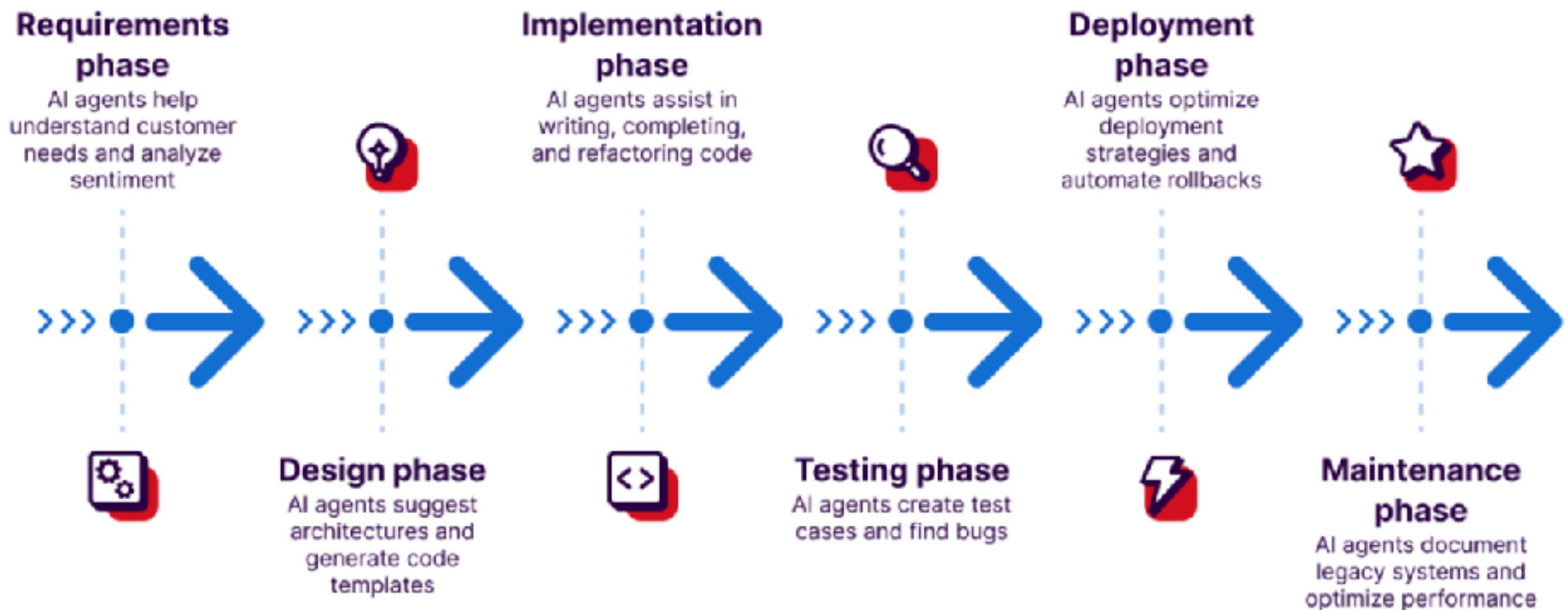
Working with AI Agent !!



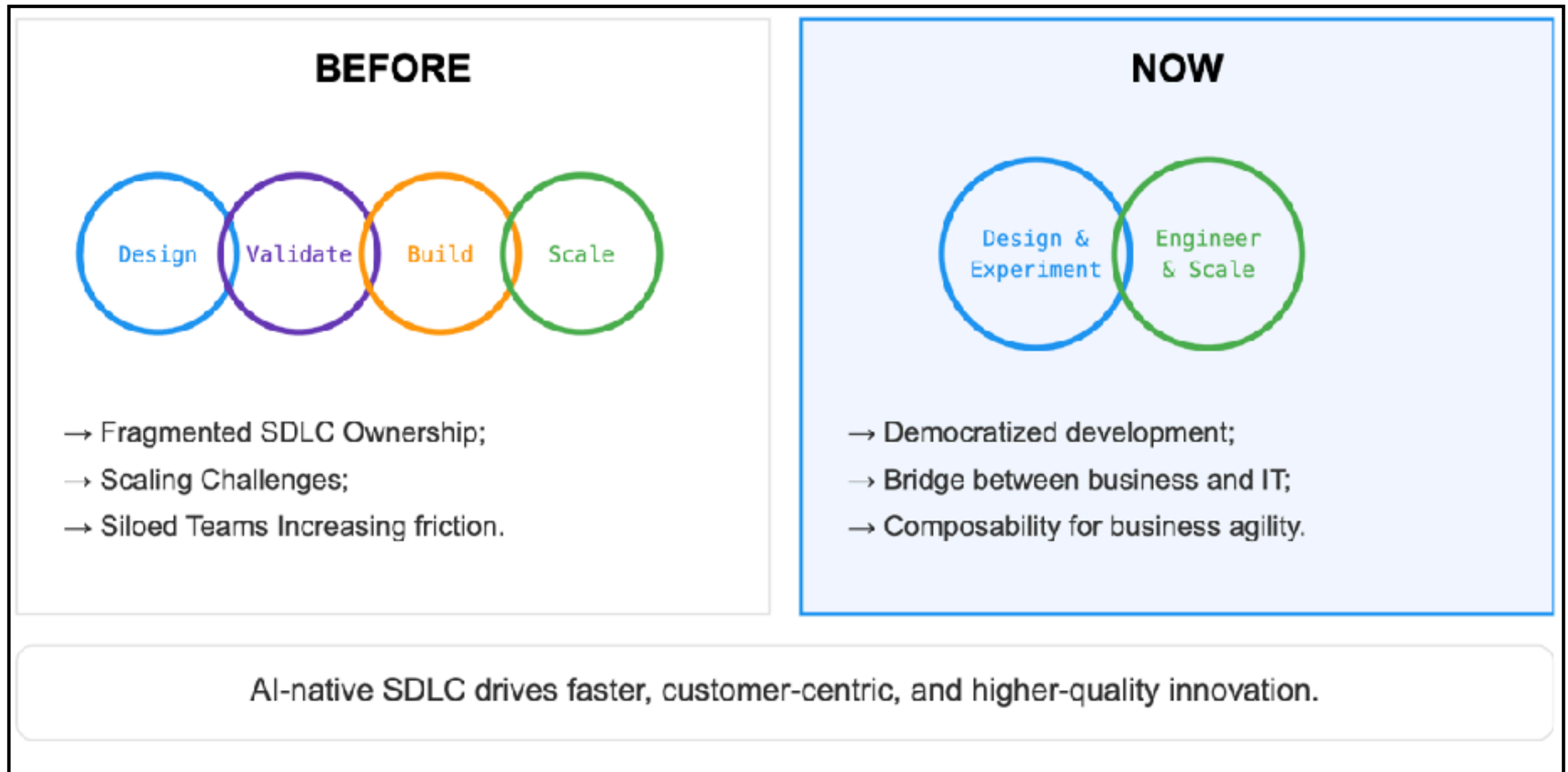
AI is the new normal in software development



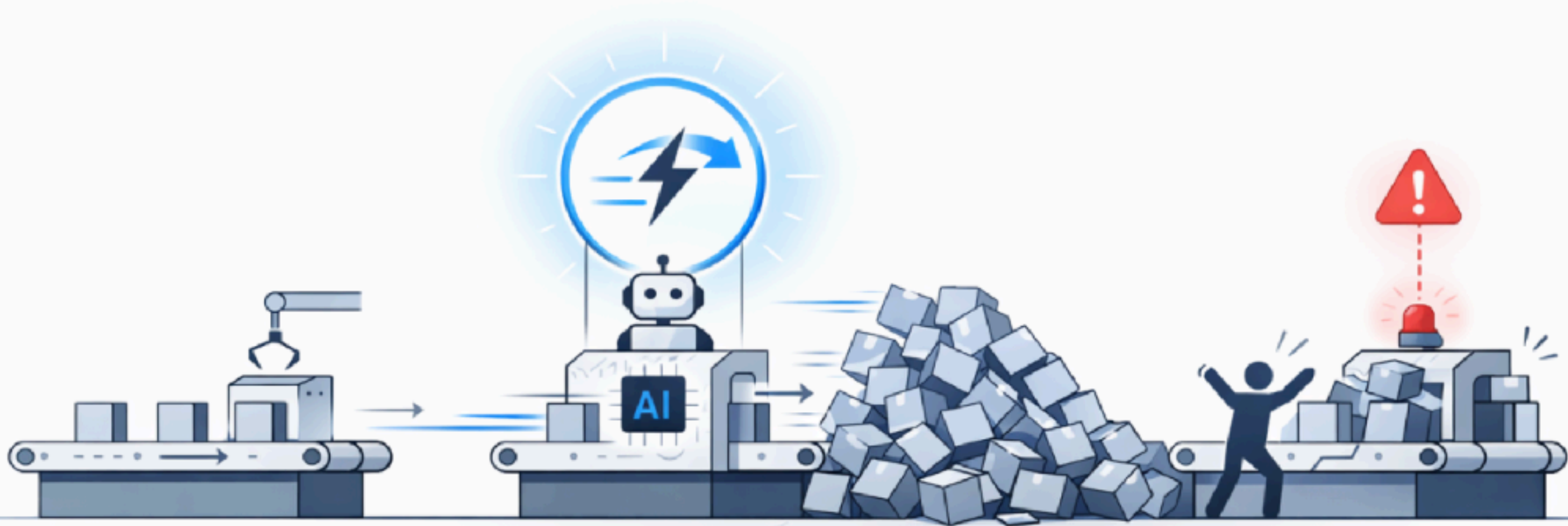
AI in Software Development



AI in Software Development

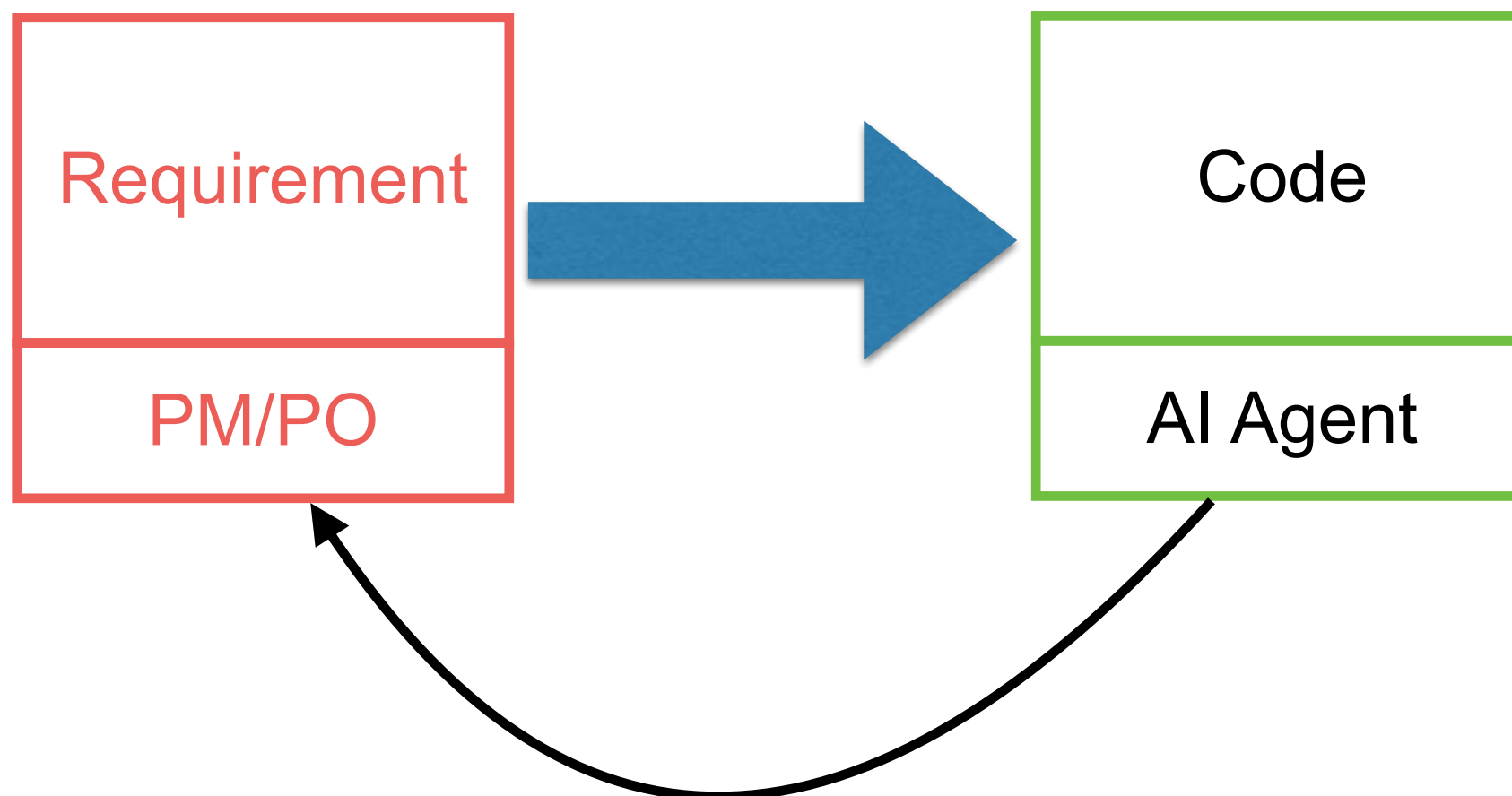


Invest in Real Bottleneck ?



New Bottleneck ?

Decide what to **build**, not code



Get feedback faster

<https://www.theaiopportunities.com/p/is-ai-slowng-down-the-new-bottleneck>



DORA



State of AI-assisted Software Development

Google Cloud

2025

<https://cloud.google.com/blog/products/ai-machine-learning/announcing-the-2025-dora-report>



AI and Productivity ?

AI Adoption

90% of software development professionals are now using AI at work

Productivity

80% report that AI has increased their productivity

Reliance

65% are relying on AI at least a “moderate amount” at work

Code Quality

59% of survey respondents say that AI has positively impacted their code quality

<https://blog.google/innovation-and-ai/technology/developers-tools/dora-report-2025/>

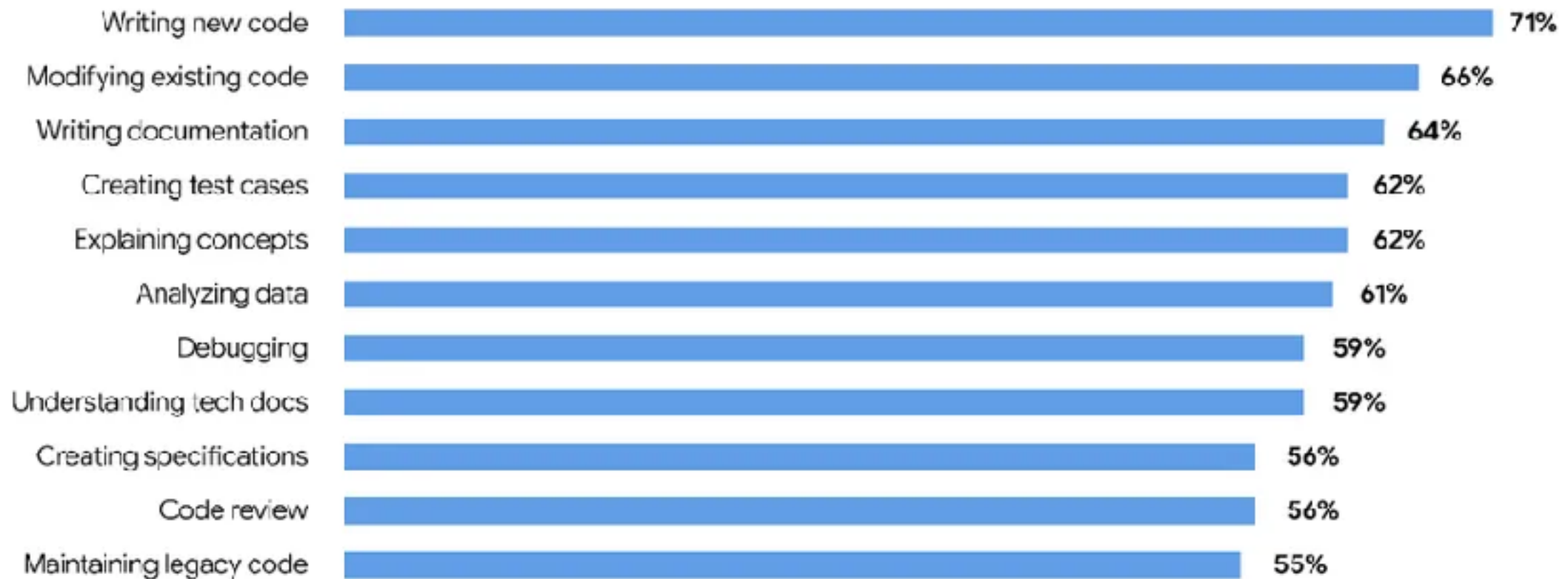


Productivity ?

Reliance on AI by task

DORA
2025

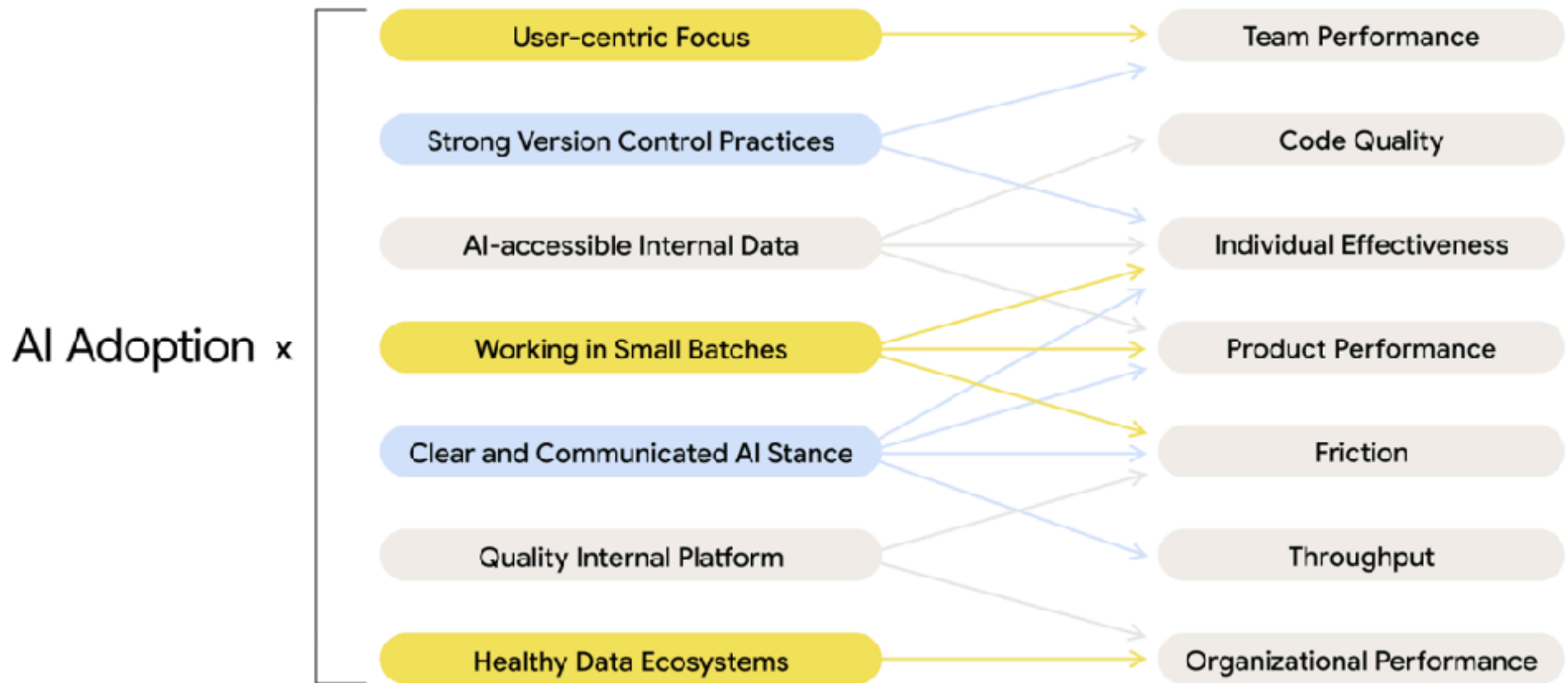
Percentage of task performers who use AI



<https://blog.google/innovation-and-ai/technology/developers-tools/dora-report-2025/>



DORA AI Capabilities Model



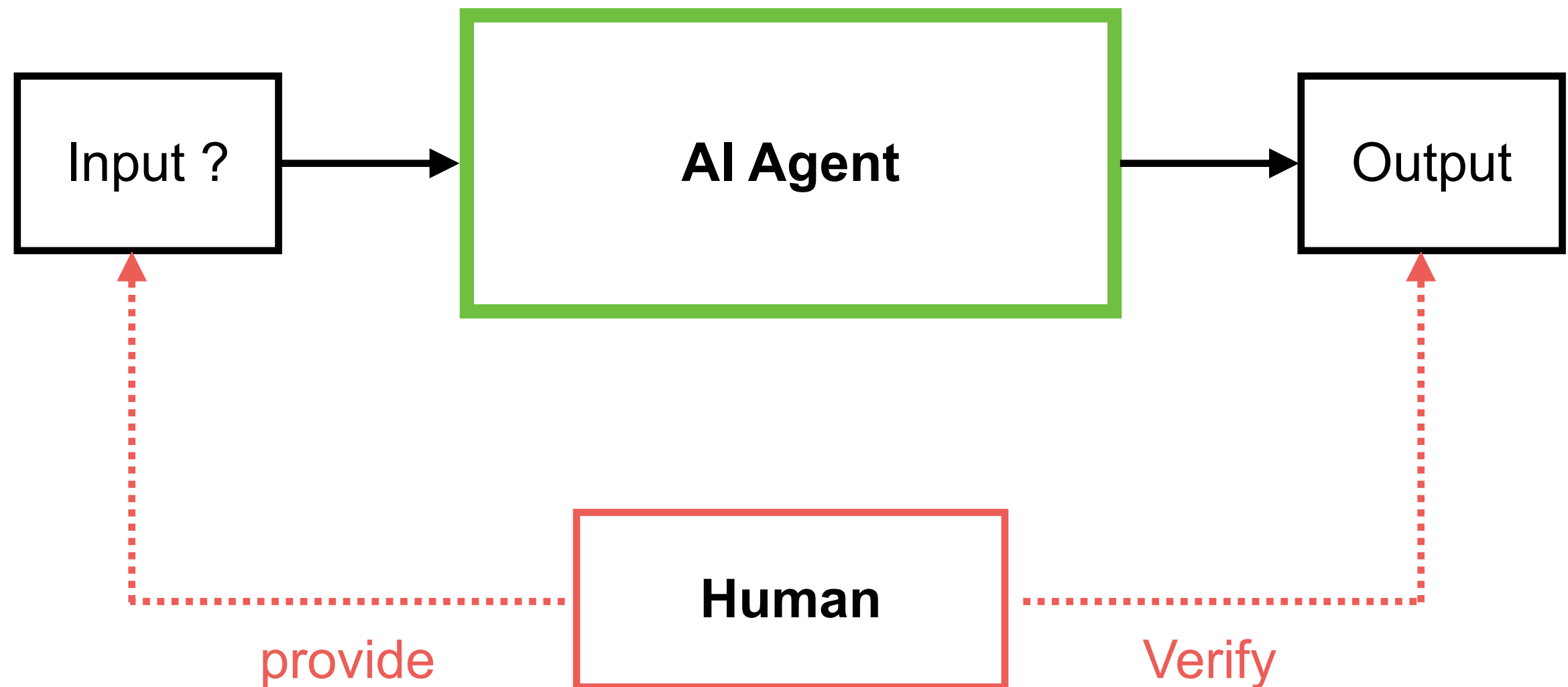
<https://cloud.google.com/blog/products/ai-machine-learning/announcing-the-2025-dora-report>



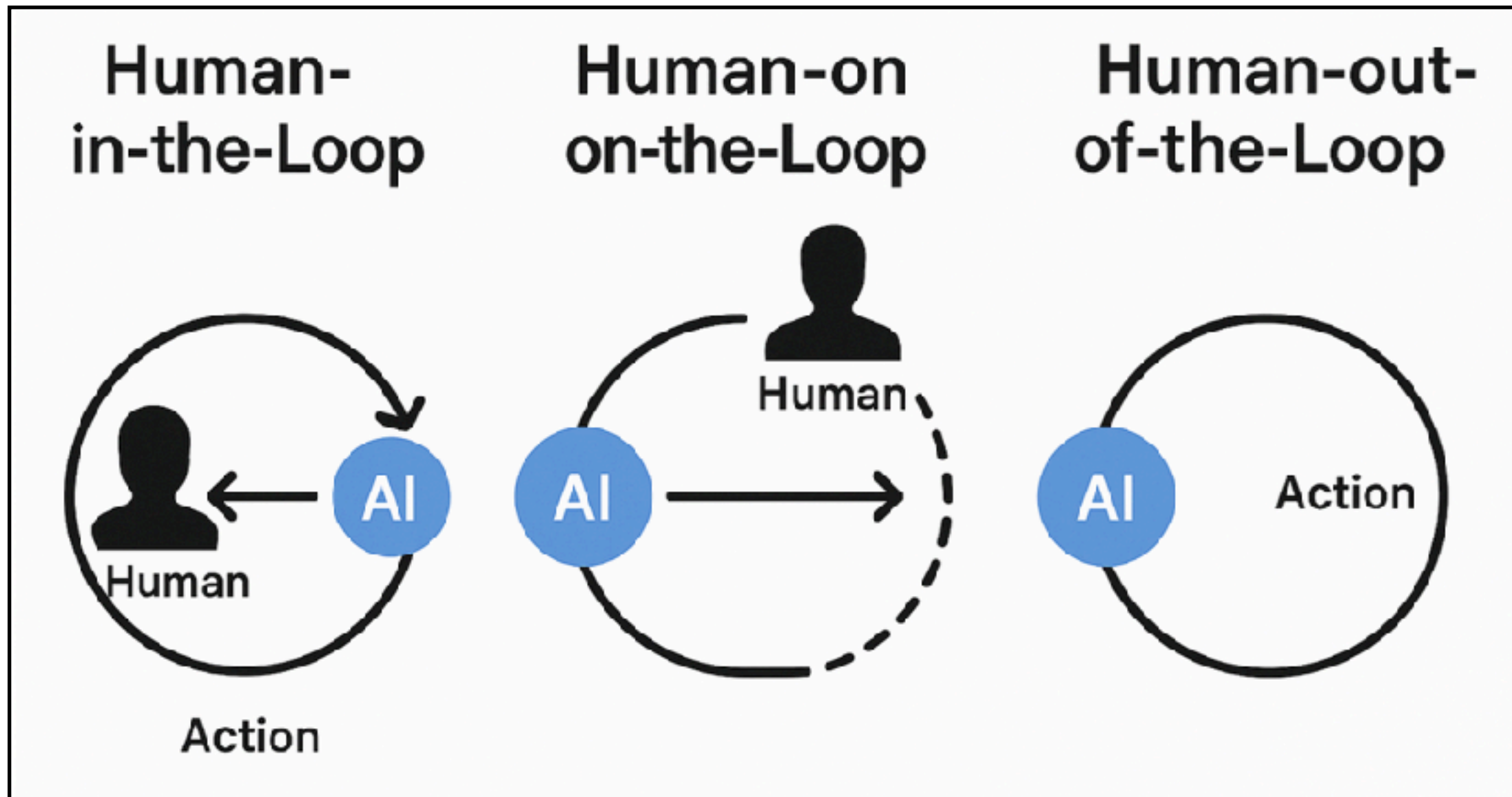
AI Agent for Coding



AI Agent + Human **in/on** the Loop



AI Agent with/without Human

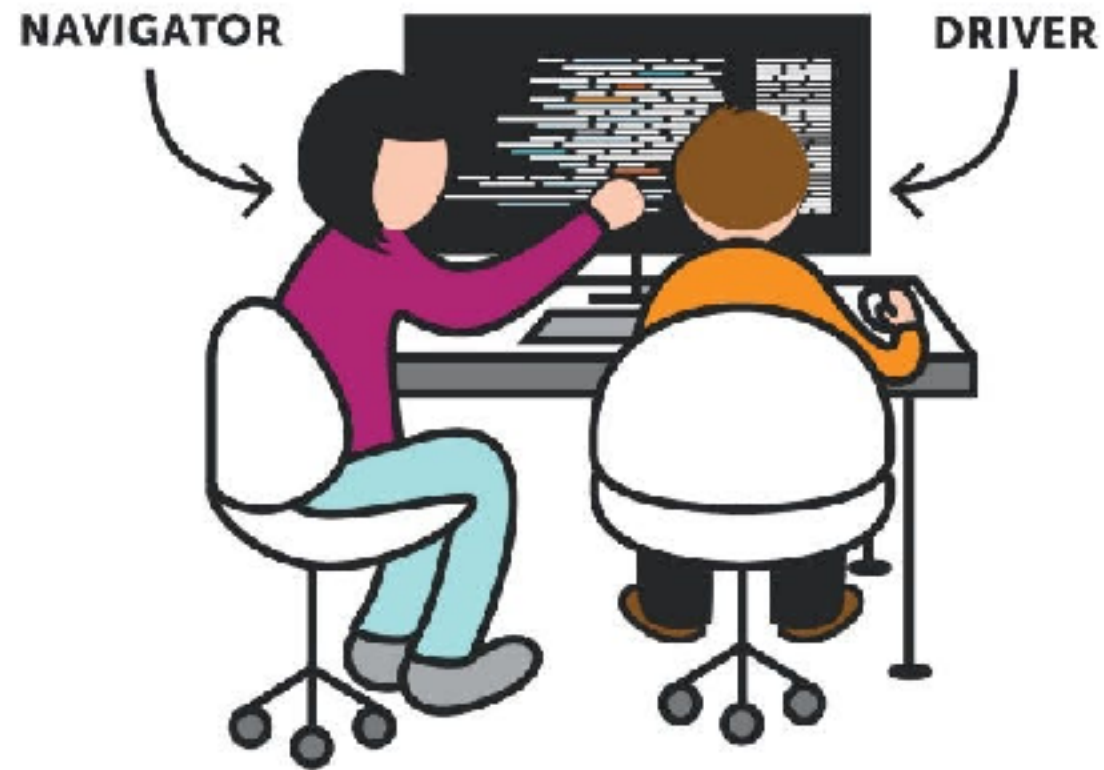


AI draft

Human craft



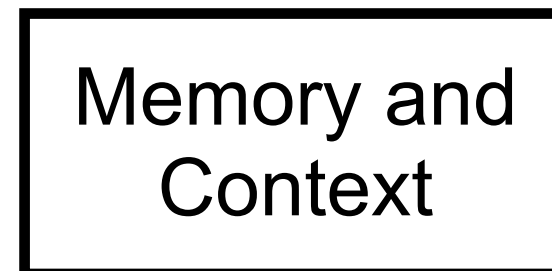
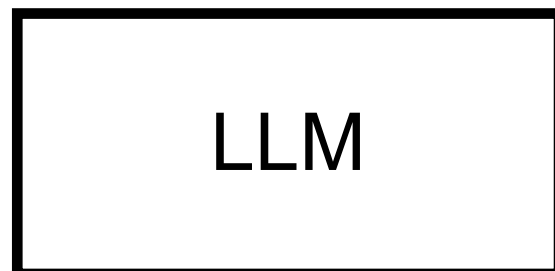
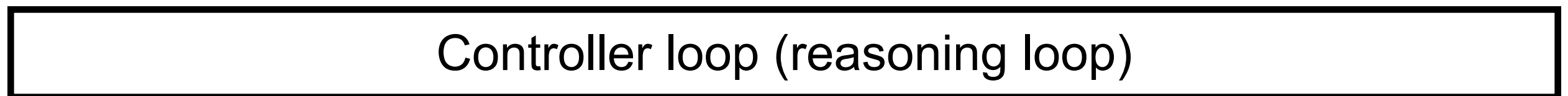
PAIR PROGRAMMING



Reviews, Tests and Understands
with experience

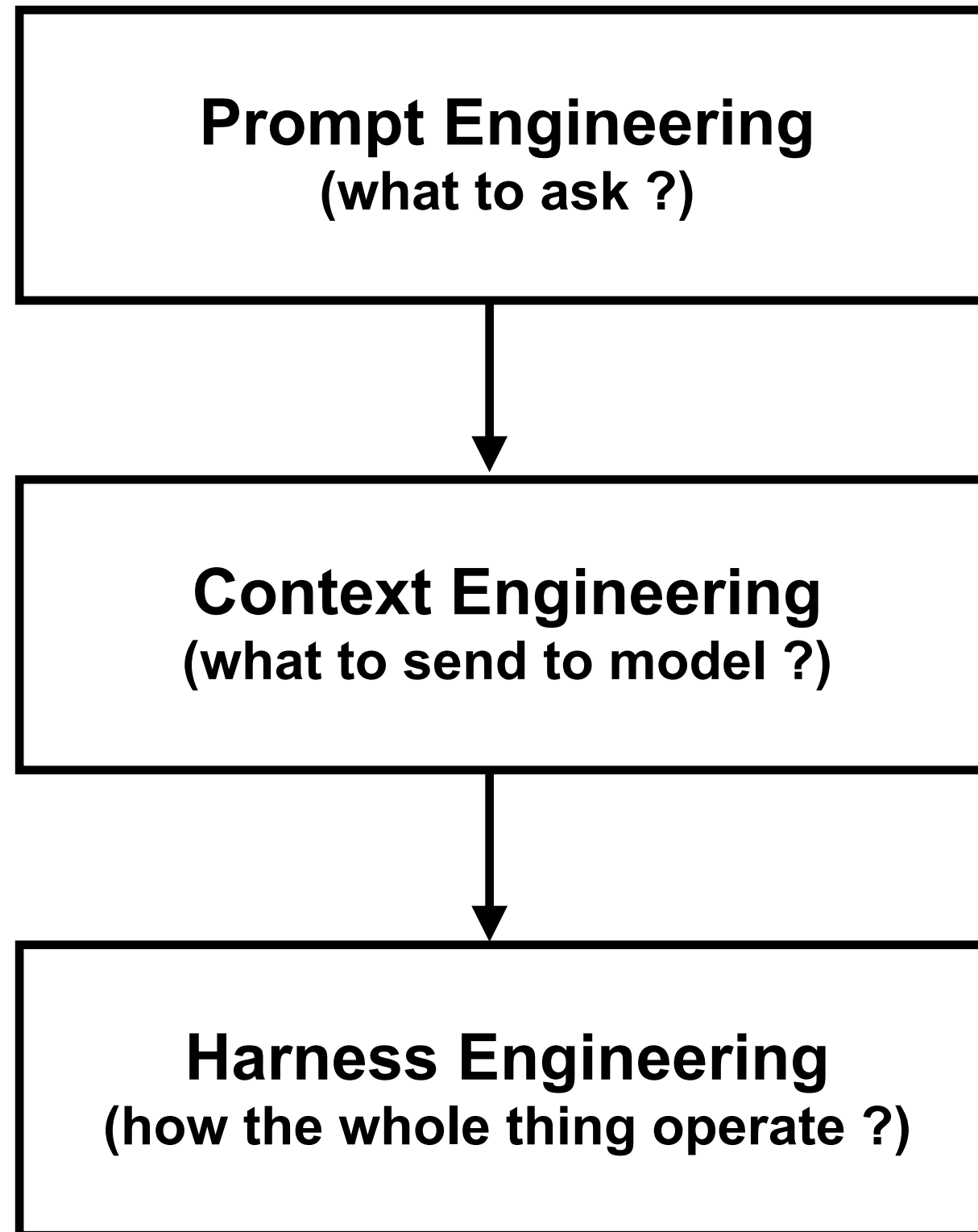


AI Agent for Coding



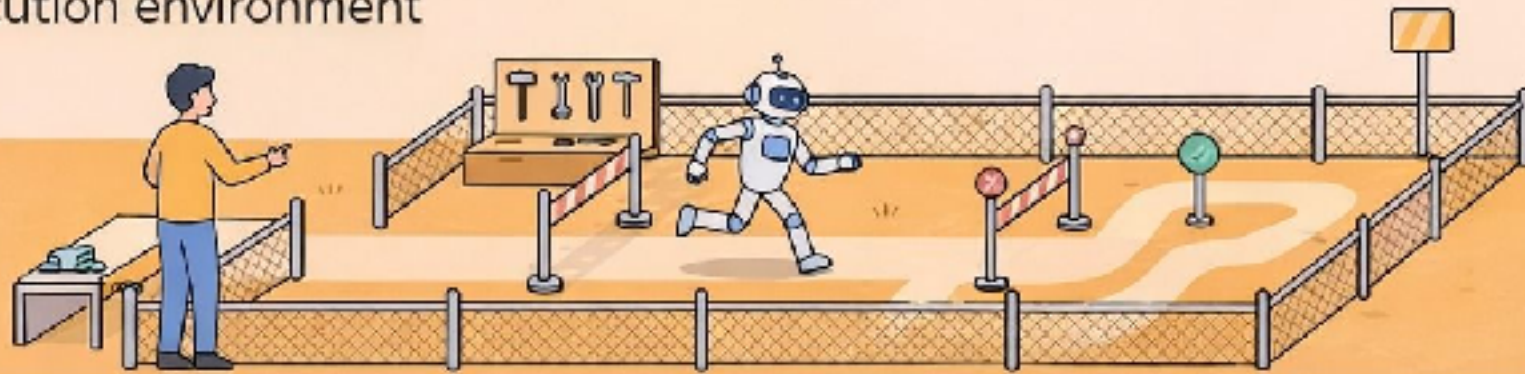
How to work with AI Agent ?





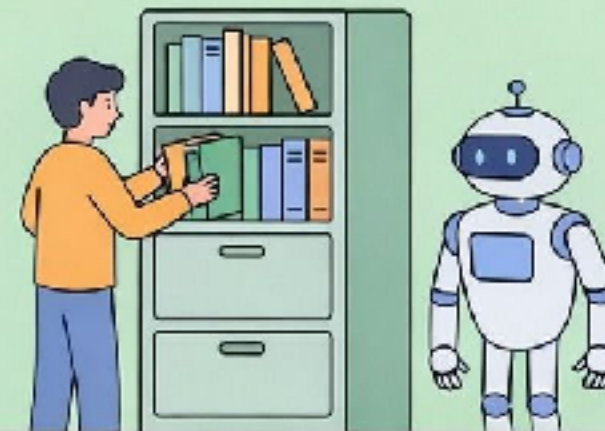
Harness Engineering — Build the World

Design the execution environment



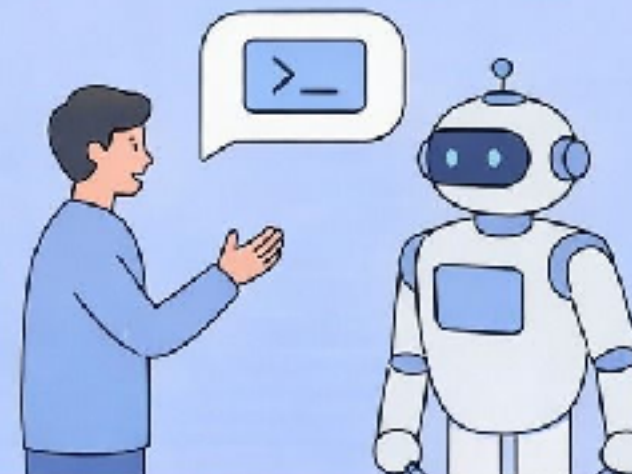
Context Engineering — What to See

Manage visible information



Prompt Engineering — What to Say

Optimize single-turn interactions



<https://milvus.io/blog/harness-engineering-ai-agents.md>



2 Background: From Weights to Context to Harness

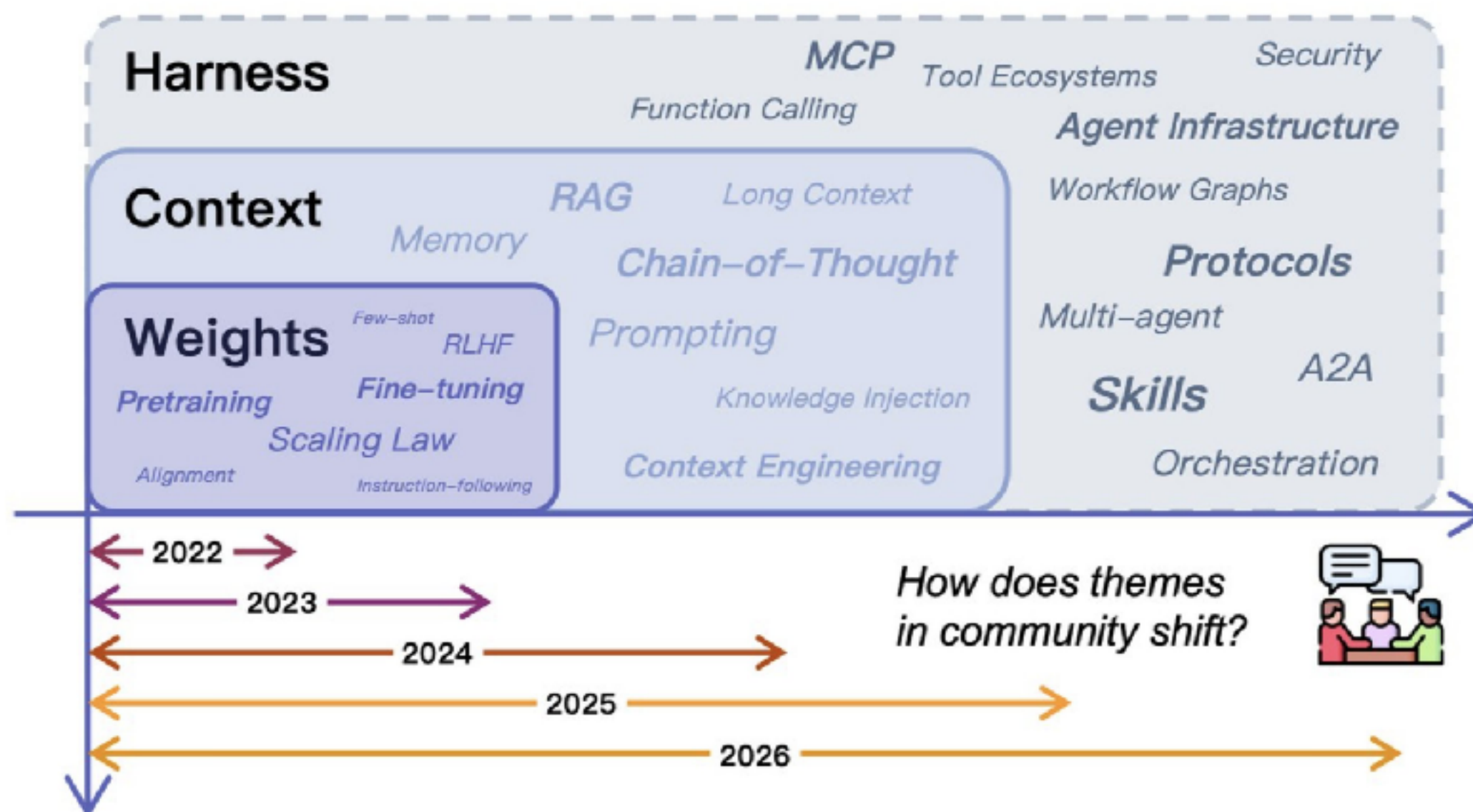
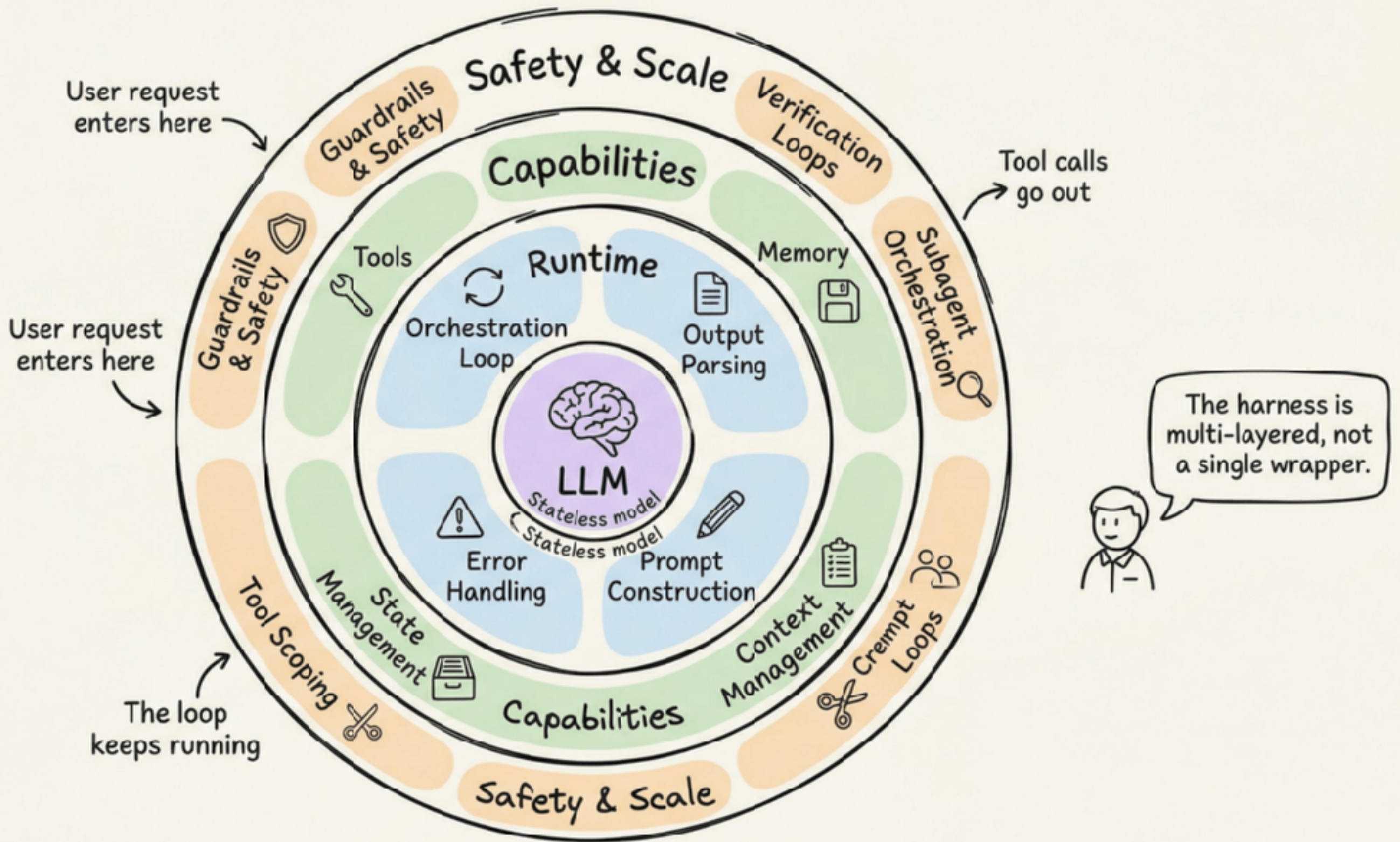


Figure 2 Community theme evolution across three capability layers. The stacked layers—Weights, Context, and Harness—show how the center of gravity in the LLM agent community has shifted outward over time, from parametric knowledge and prompting toward harness-level infrastructure such as tool ecosystems, protocols, skills, and multi-agent orchestration.

<https://arxiv.org/pdf/2604.08224>





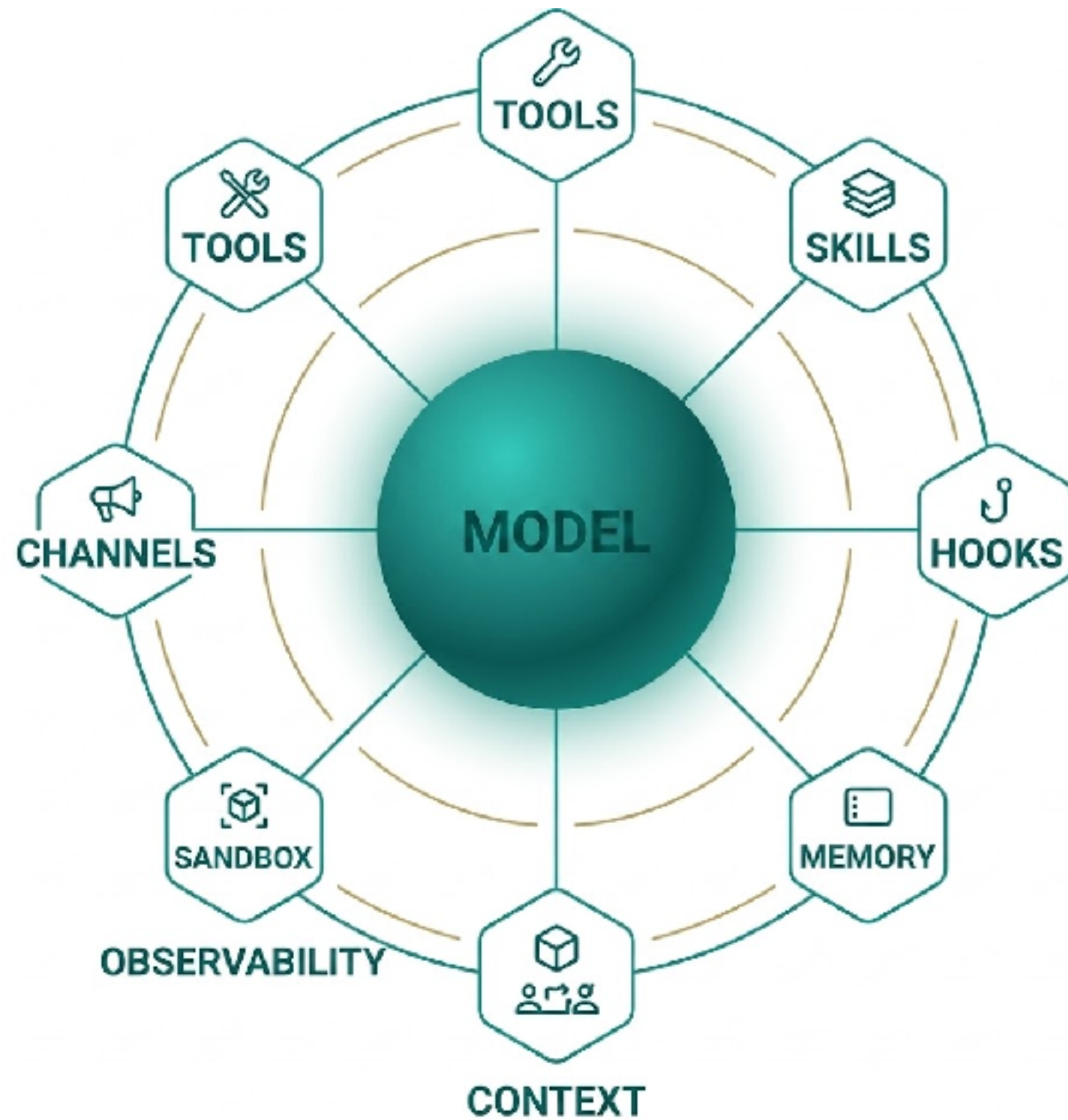
<https://blog.dailydoseofds.com/p/evolution-of-agent-landscape-from>



AI Agent = LLM + Harness

Harness = AI Agent - LLM





Workflows with AI Agent ?



Workflows ?

Application level

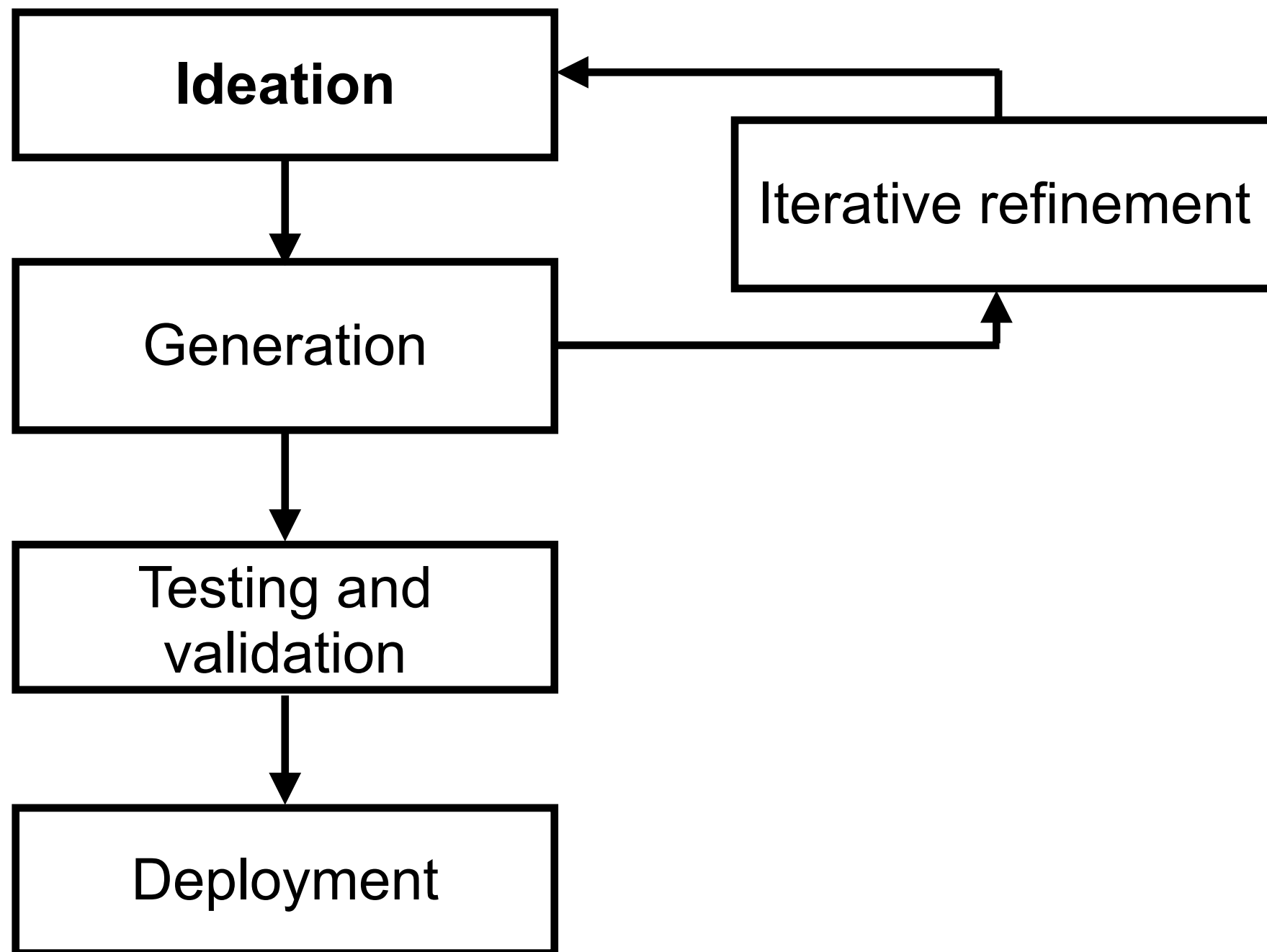
Code level

<https://cloud.google.com/discover/what-is-vibe-coding>



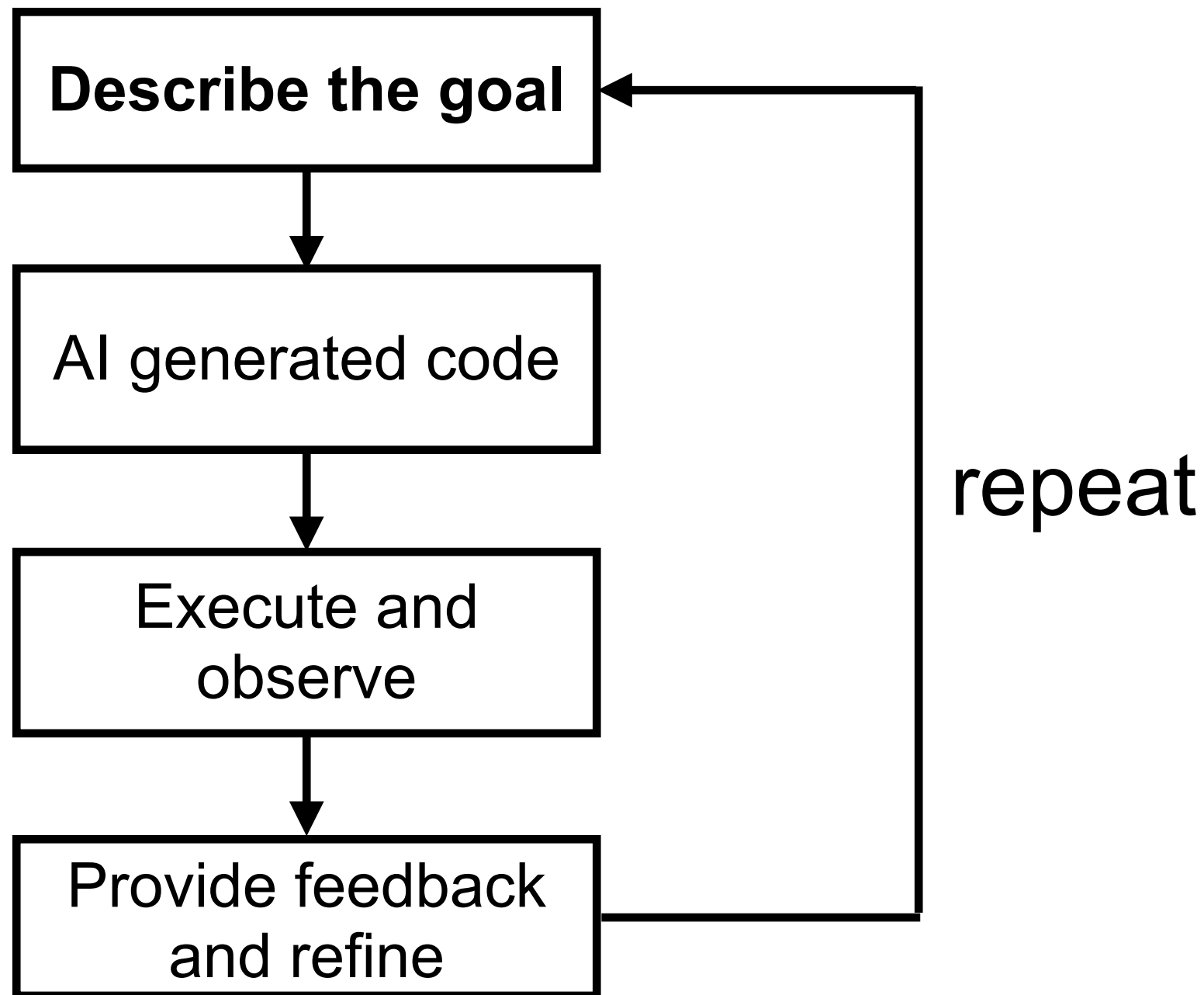
Application level workflow

Talking about **high-level idea** from concept to deployment



Code level workflow

Conversational loop to create a **specific code**

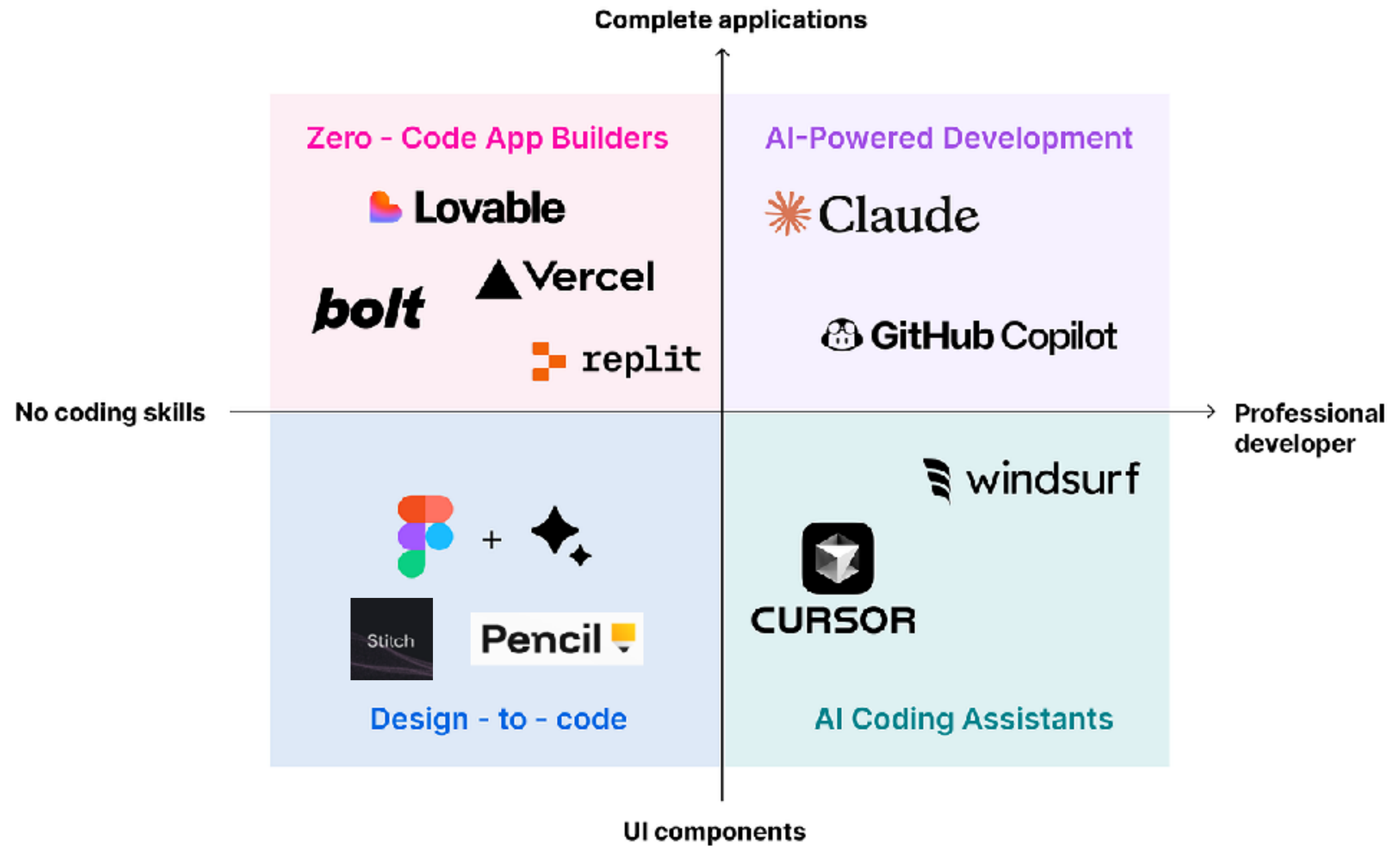


Tools ?



AI Agent for Coding !!

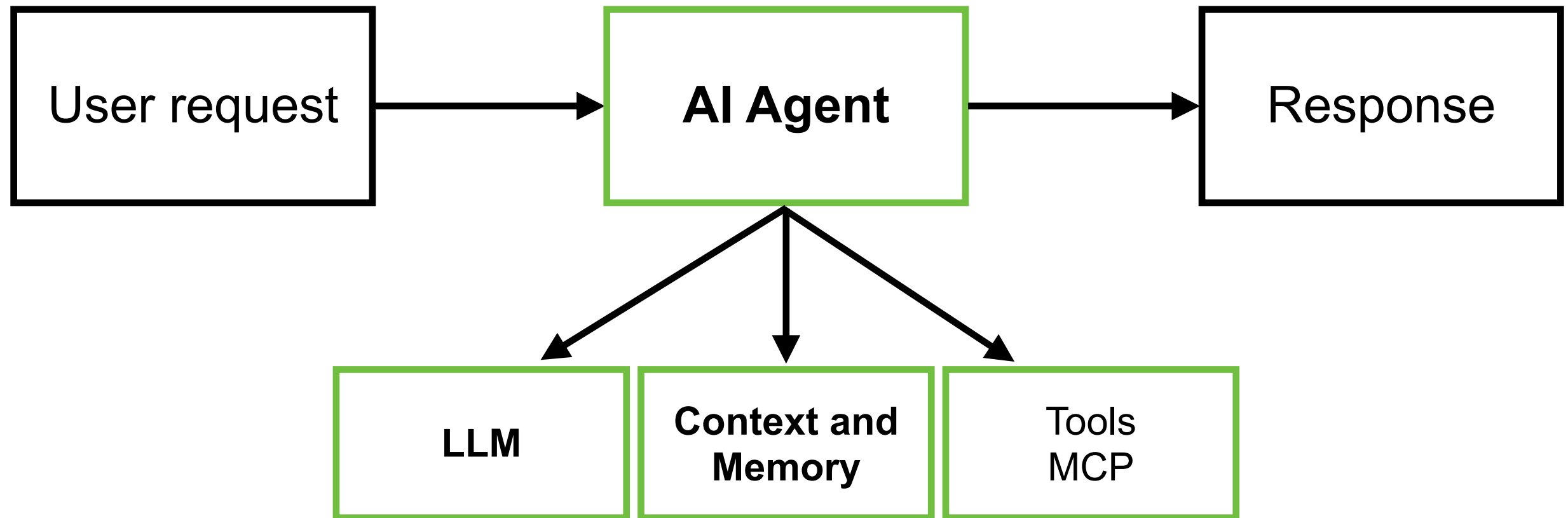




Before start ...



AI Agent for Coding



How to manage context or knowledge for Agent ?



Token is new oil !!
Token is expensive !!



LLM Model :: GPT 5.5

**GPT-5.5** Default Compare Try in Playground

A new class of intelligence for coding and professional work.

REASONING

Highest

SPEED

Fast

PRICE
\$5 - \$30
Input - Output

INPUT
   
Text, image

OUTPUT
   
Text

GPT-5.5 is our newest frontier model for the most complex professional work. Learn more in our [latest model guide](#). Reasoning effort supports: none, low, medium (default), high and xhigh.

- ✦ 1,050,000 context window
- ➡ 128,000 max output tokens
- 📅 Dec 01, 2025 knowledge cutoff
- 🗨 Reasoning token support

Pricing

Pricing is based on the number of tokens used, or other metrics based on the model type. For tool-specific models, like search and computer use, there's a fee per tool call. See details in the [pricing page](#).

Text tokens Per 1M tokens - Batch API price ☐

Input \$5.00	Cached input \$0.50	Output \$30.00
------------------------	-------------------------------	--------------------------

<https://developers.openai.com/api/docs/models/gpt-5.5>



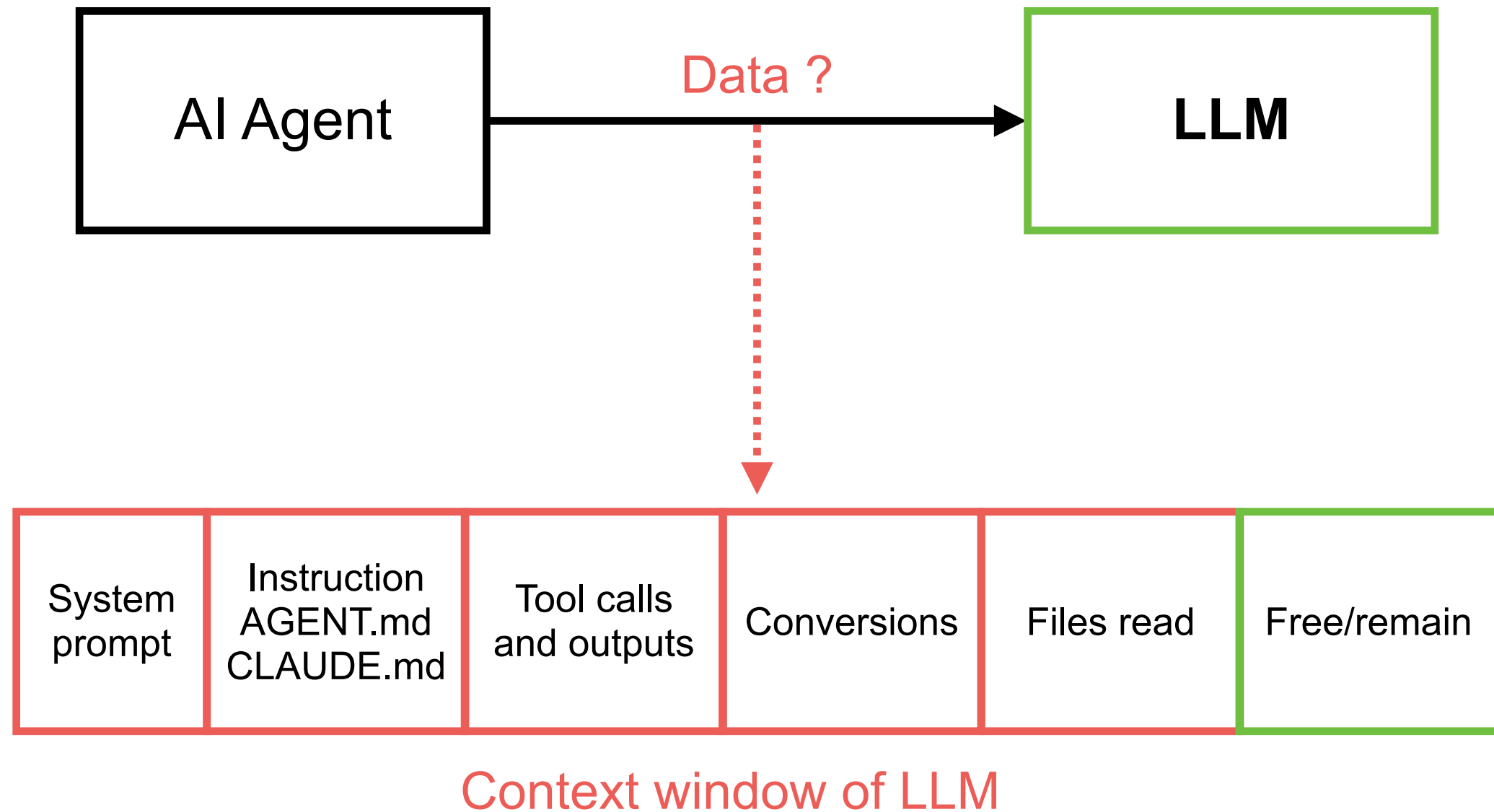
LLM Model :: Anthropic

Model	Base Input Tokens	5m Cache Writes	1h Cache Writes	Cache Hits & Refreshes	Output Tokens
Claude Opus 4.7	\$5 / MTok	\$6.25 / MTok	\$10 / MTok	\$0.50 / MTok	\$25 / MTok
Claude Opus 4.6	\$5 / MTok	\$6.25 / MTok	\$10 / MTok	\$0.50 / MTok	\$25 / MTok
Claude Opus 4.5	\$5 / MTok	\$6.25 / MTok	\$10 / MTok	\$0.50 / MTok	\$25 / MTok
Claude Opus 4.1	\$15 / MTok	\$18.75 / MTok	\$30 / MTok	\$1.50 / MTok	\$75 / MTok
Claude Opus 4	\$15 / MTok	\$18.75 / MTok	\$30 / MTok	\$1.50 / MTok	\$75 / MTok
Claude Sonnet 4.6	\$3 / MTok	\$3.75 / MTok	\$6 / MTok	\$0.30 / MTok	\$15 / MTok
Claude Sonnet 4.5	\$3 / MTok	\$3.75 / MTok	\$6 / MTok	\$0.30 / MTok	\$15 / MTok
Claude Sonnet 4	\$3 / MTok	\$3.75 / MTok	\$6 / MTok	\$0.30 / MTok	\$15 / MTok

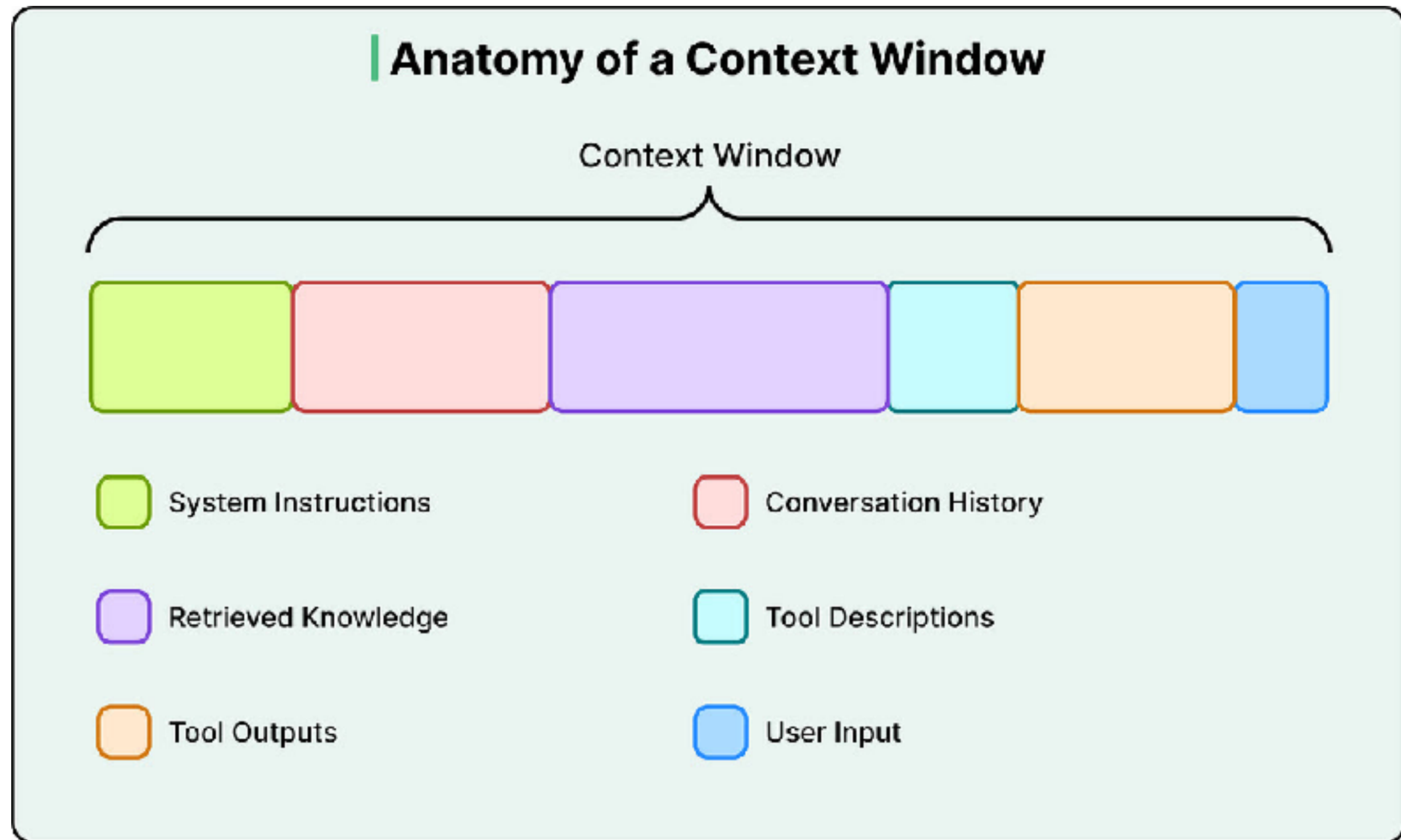
<https://platform.claude.com/docs/en/about-claude/pricing>



Context Window ?



Anatomy of Context Window



Types of Knowledge/Context ?

Shared context

Static Knowledge

Prompt engineering
Global Instructions and rules
AGENTS.md
CLAUDE.md
DESIGN.md

Task context

Dynamic Knowledge

Memory
Graph/Vector database
MCP tools
Specific Instructions
Agent SKILLS

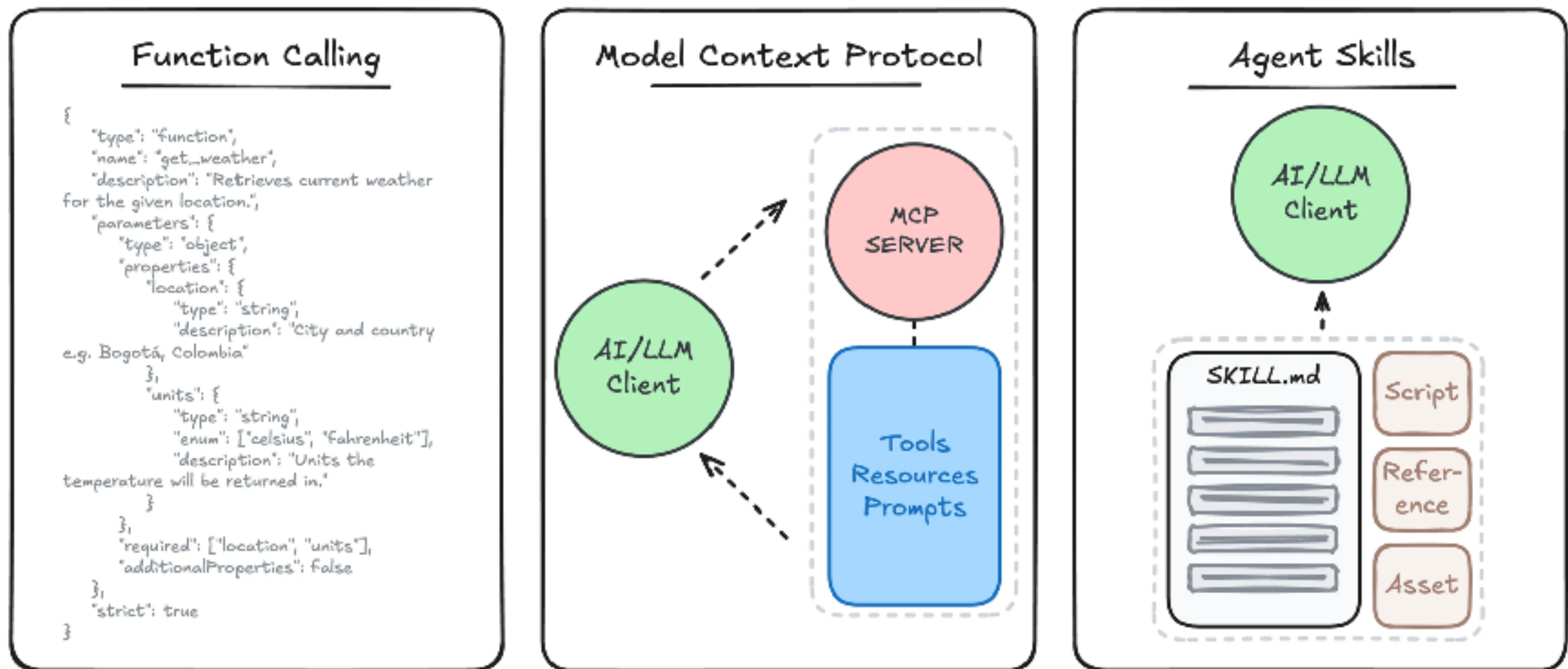
Sub-agent and Modes

Context engineering ?



Agent Skills

Standard way to give AI agent new capabilities and expertise



<https://agentskills.io/home>



How to manage context ?



Context Engineering

Right information

Right tools

Structured context



Long context == Better result ?



Context window size of models

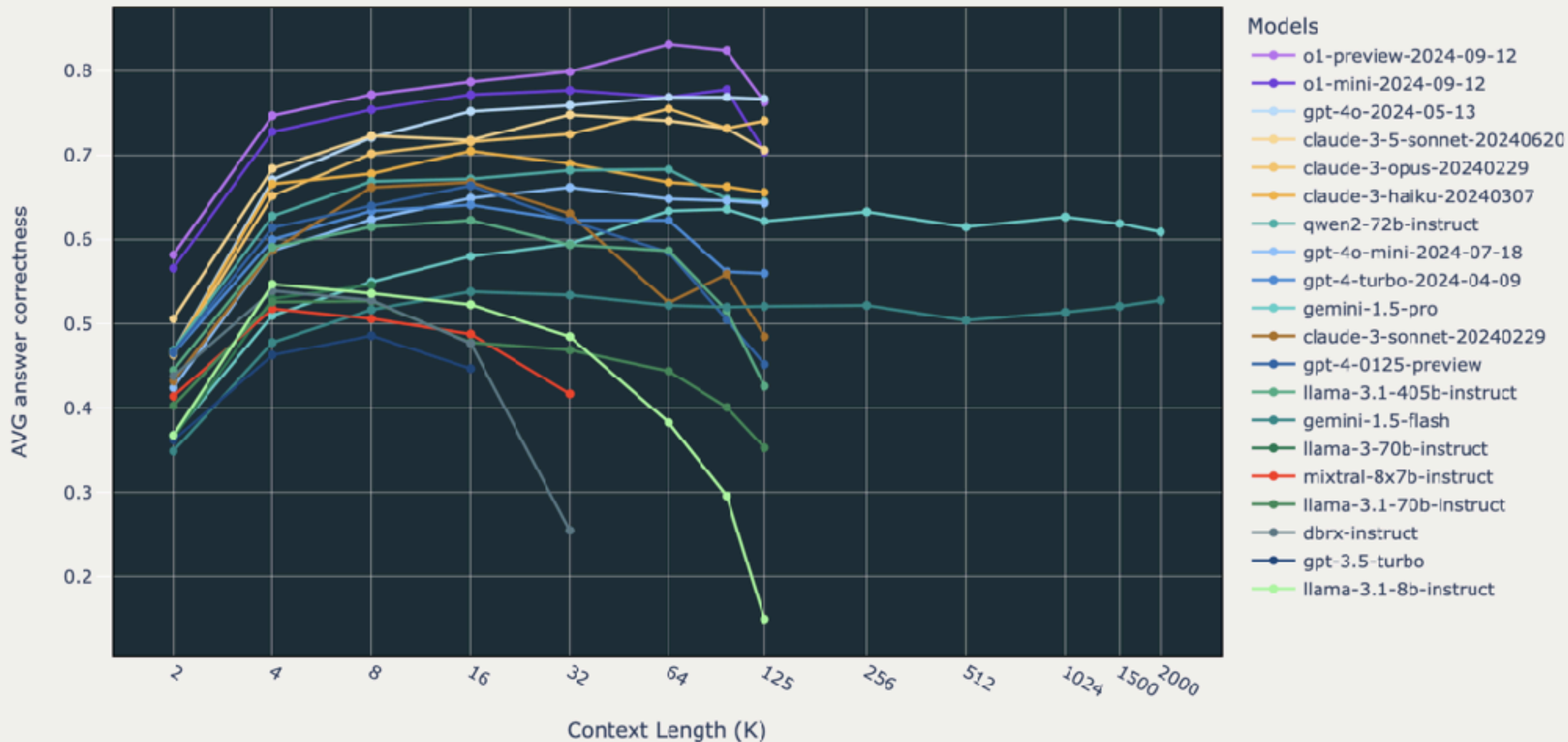
Different models may have different window size

Model name	Context window size
GPT 5.5	1M
Gemini 1.5, 2.5	1M
Llama 4	1M
Claude Opus 4.7	1M



Long context LLM is better !!

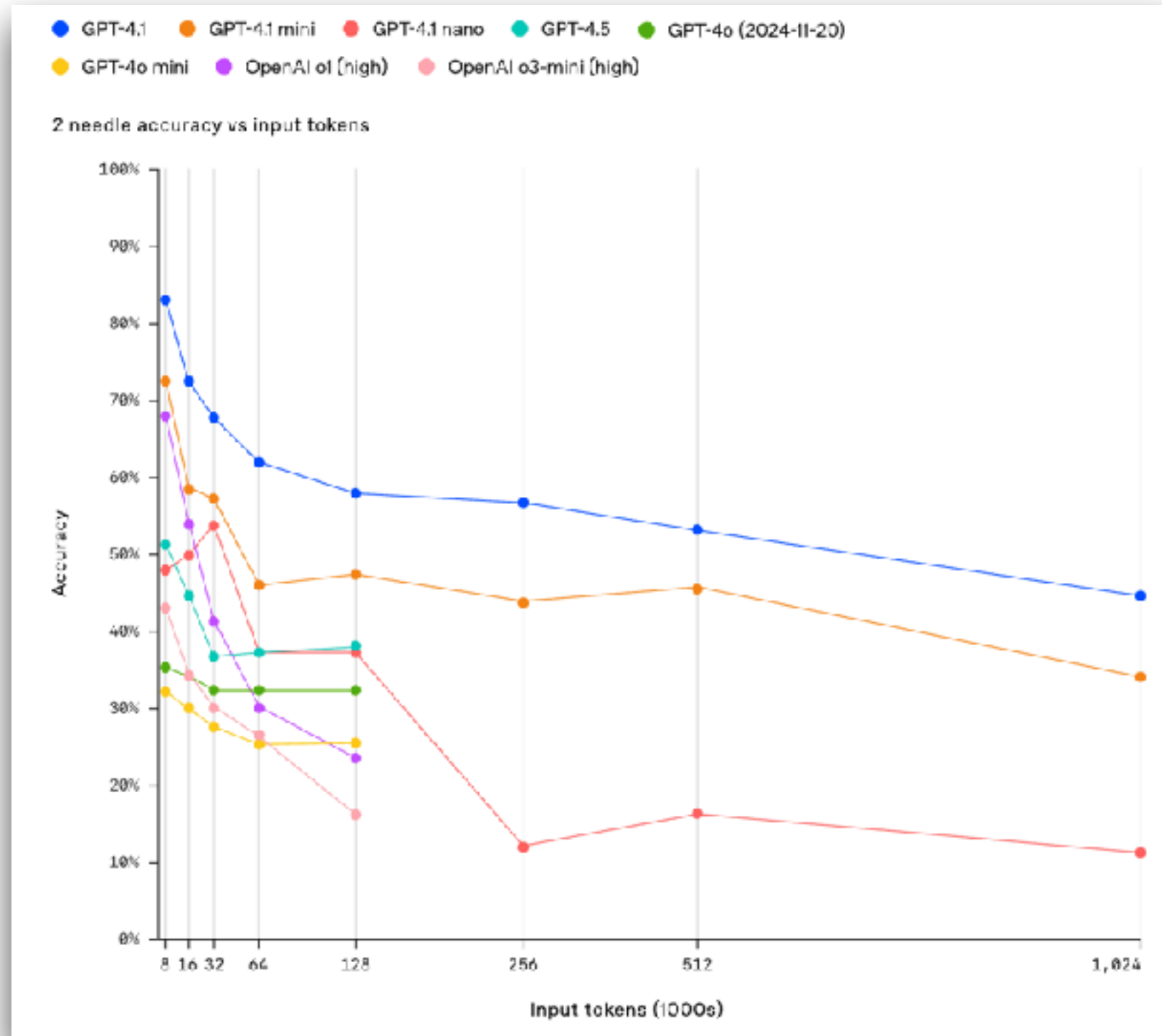
Long Context RAG Performance of LLMs



<https://arxiv.org/abs/2411.03538>



Long context LLM is better !!



<https://openai.com/index/gpt-4-1/>

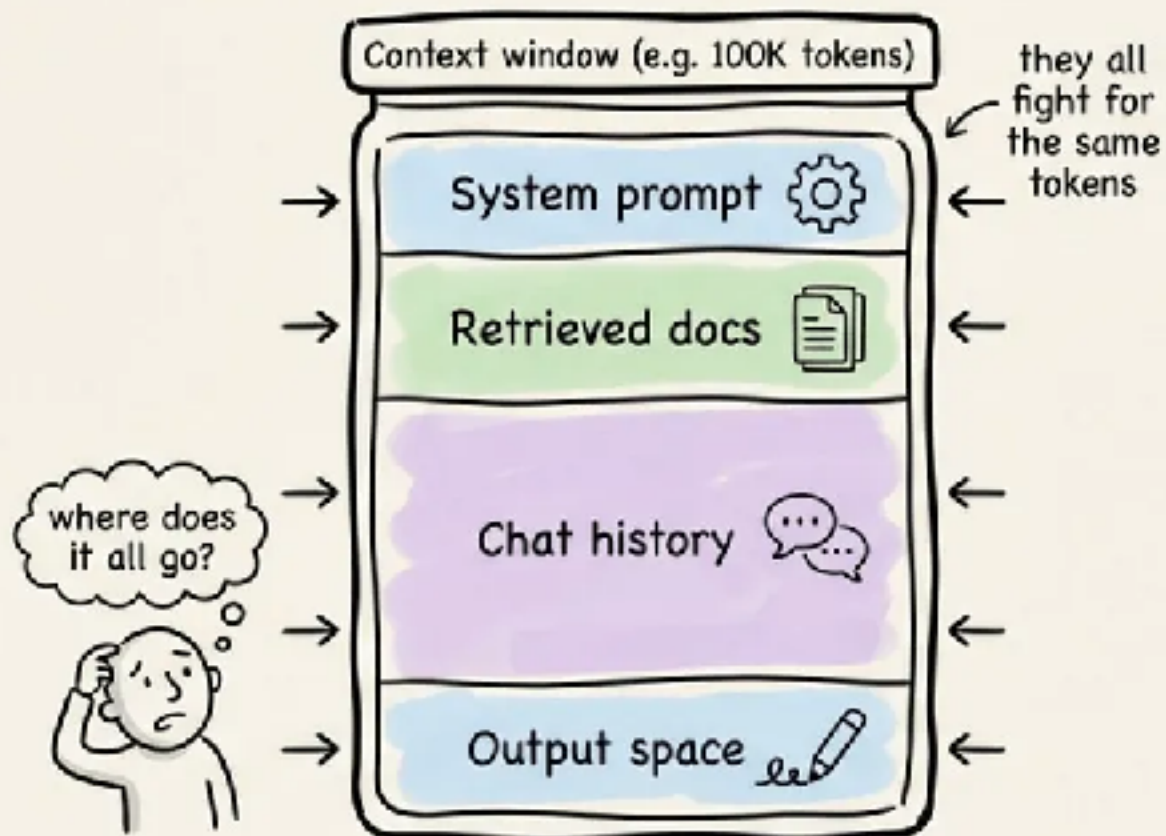


Better context == Better result ?

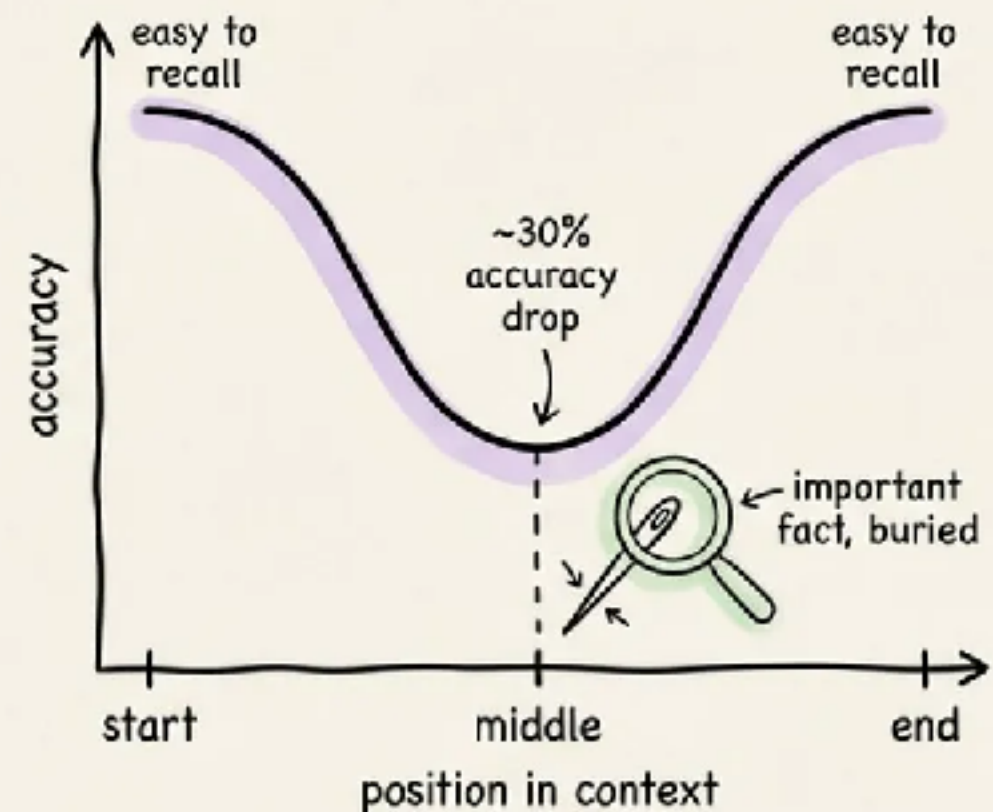


More tokens isn't more memory.

Context = a shared budget



Lost in the middle



Bigger windows don't beat structured memory.

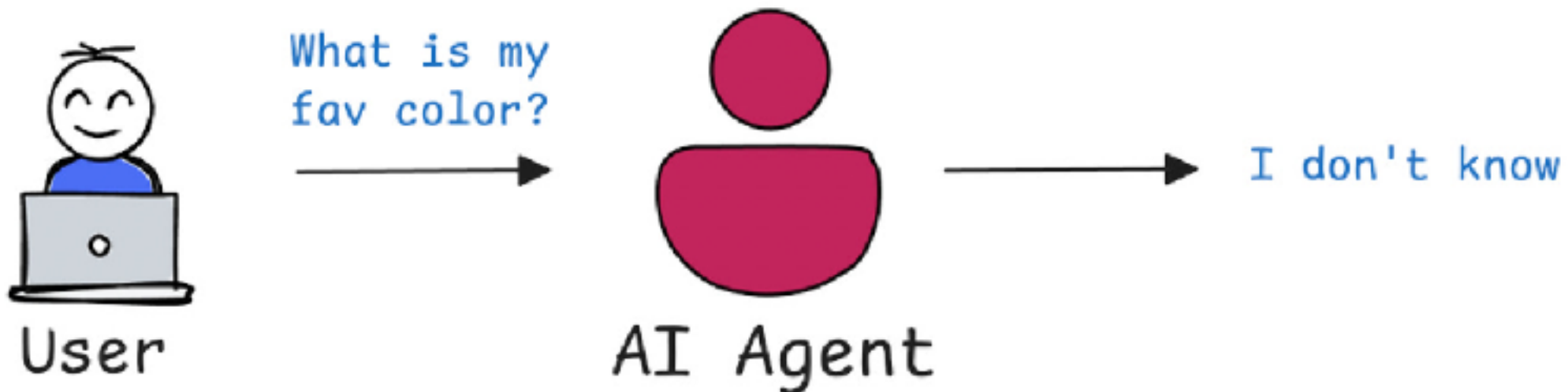
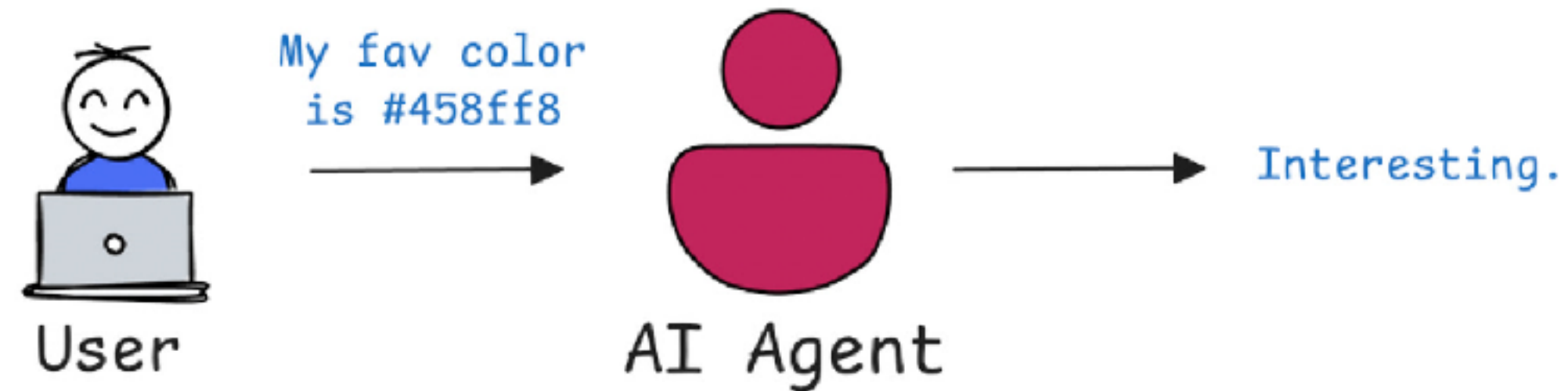
<https://blog.dailydoseofds.com/p/evolution-of-agent-landscape-from>



Agent Memory Memory Bank



Interaction without memory

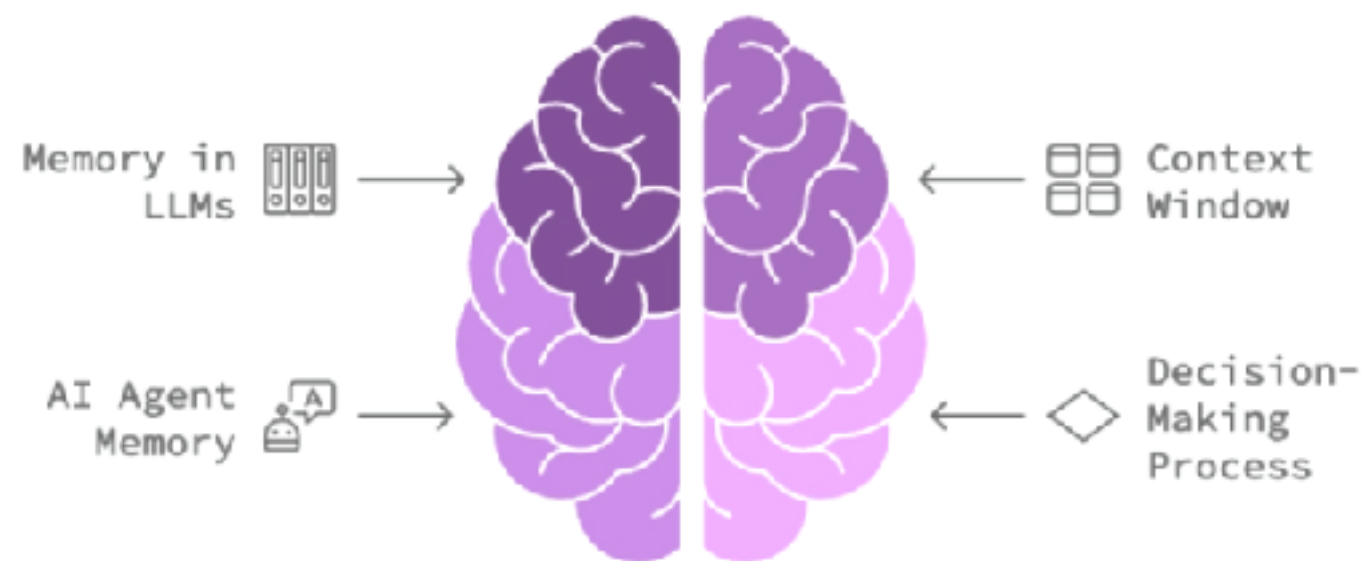


Why AI Agent memory important ?

Enable personalized and context-aware interactions

Improve decision-making and adaptability

Enhance LLM's performance and response



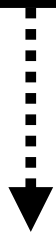
<https://www.falkordb.com/blog/ai-agents-memory-systems/>



Types of Memory

To make intelligent decisions
Agents combine different types of memory

Short-term



Current goals and steps
File system (markdown, JSON)
In-memory database

Episodic



Logs of past interactions
Success and failure

Semantic

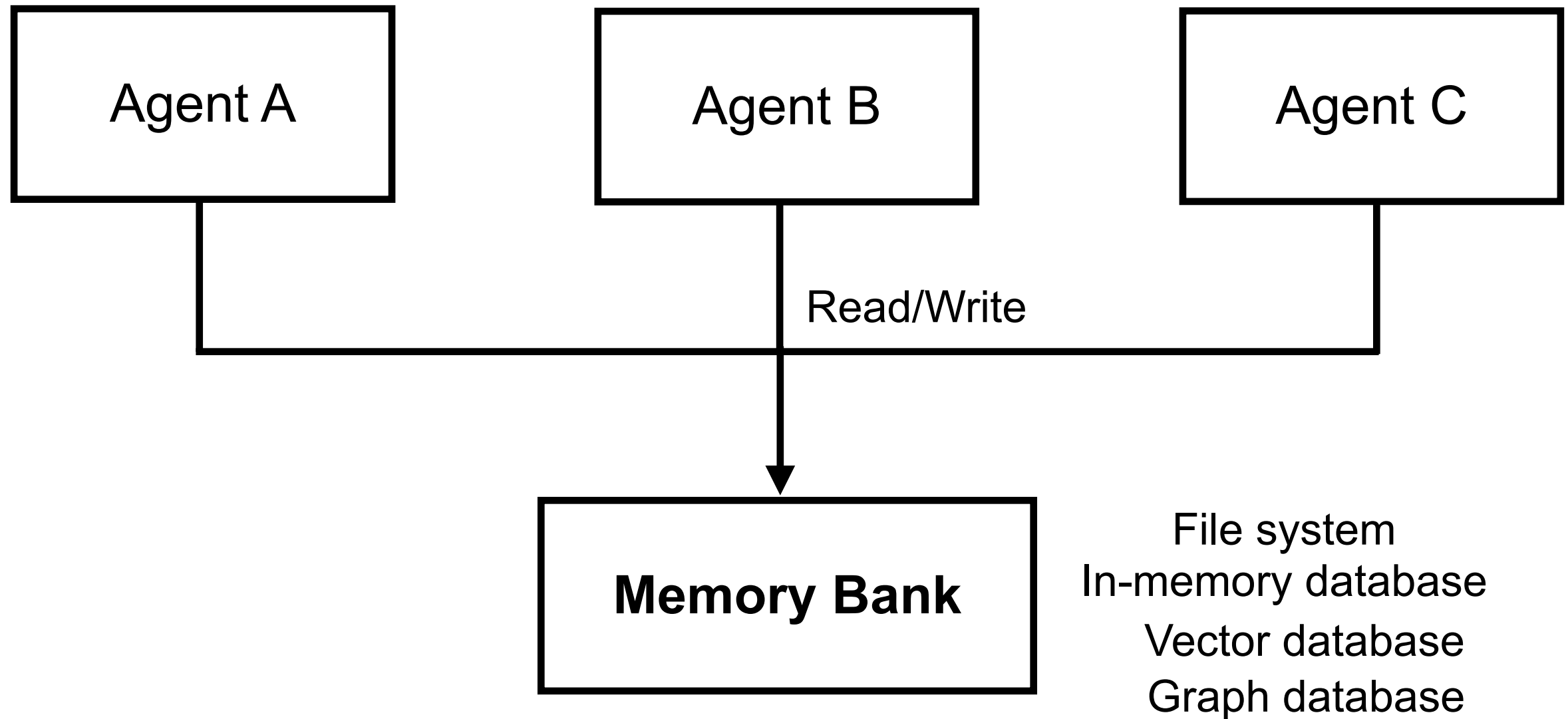


Long-term knowledge
Fact and strategies
RAG
Vector and graph database

<https://bhavishyapandit9.substack.com/p/how-memory-works-in-agentic-ai-a>

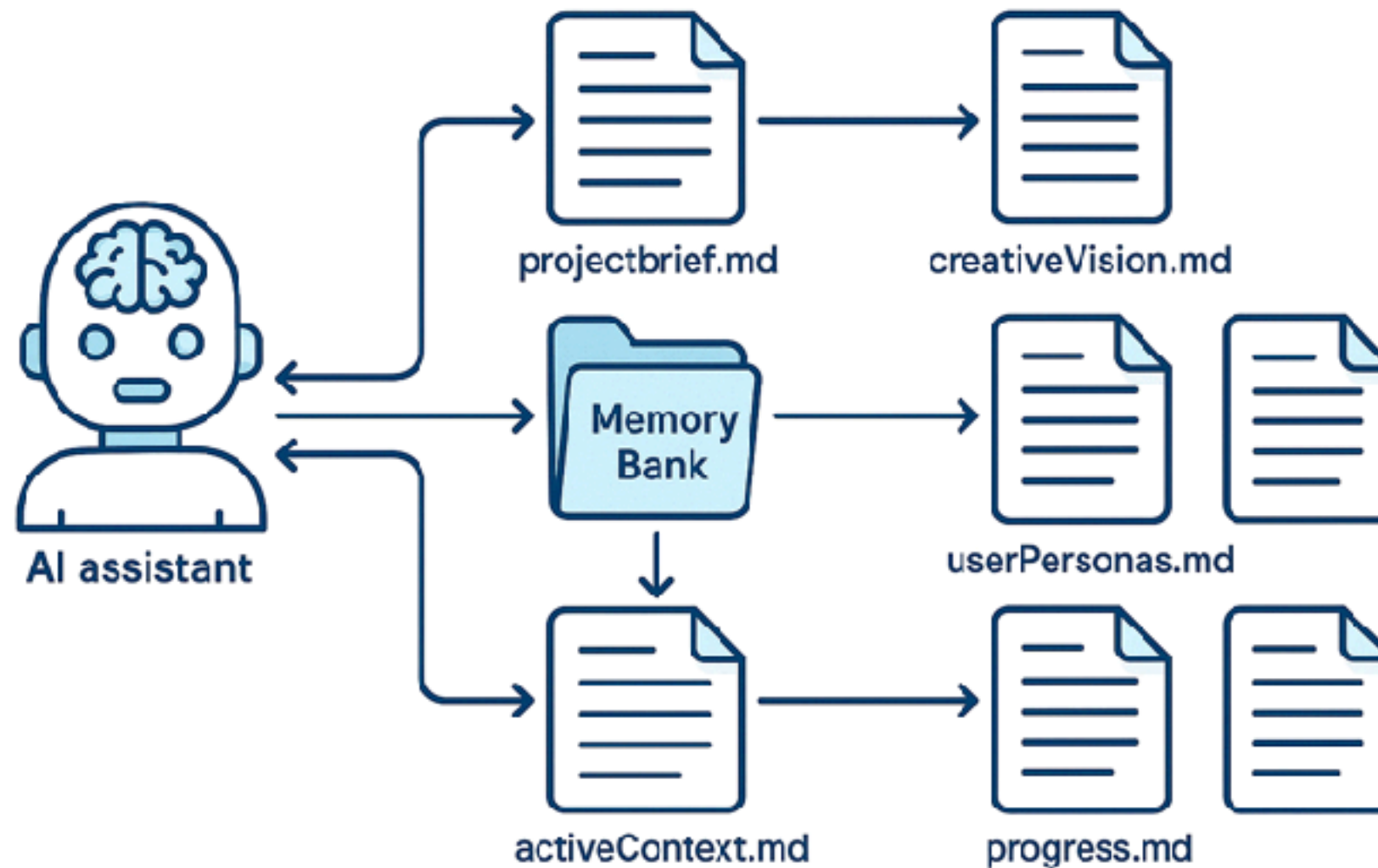


Shared memory between Agent



Cline Memory Bank

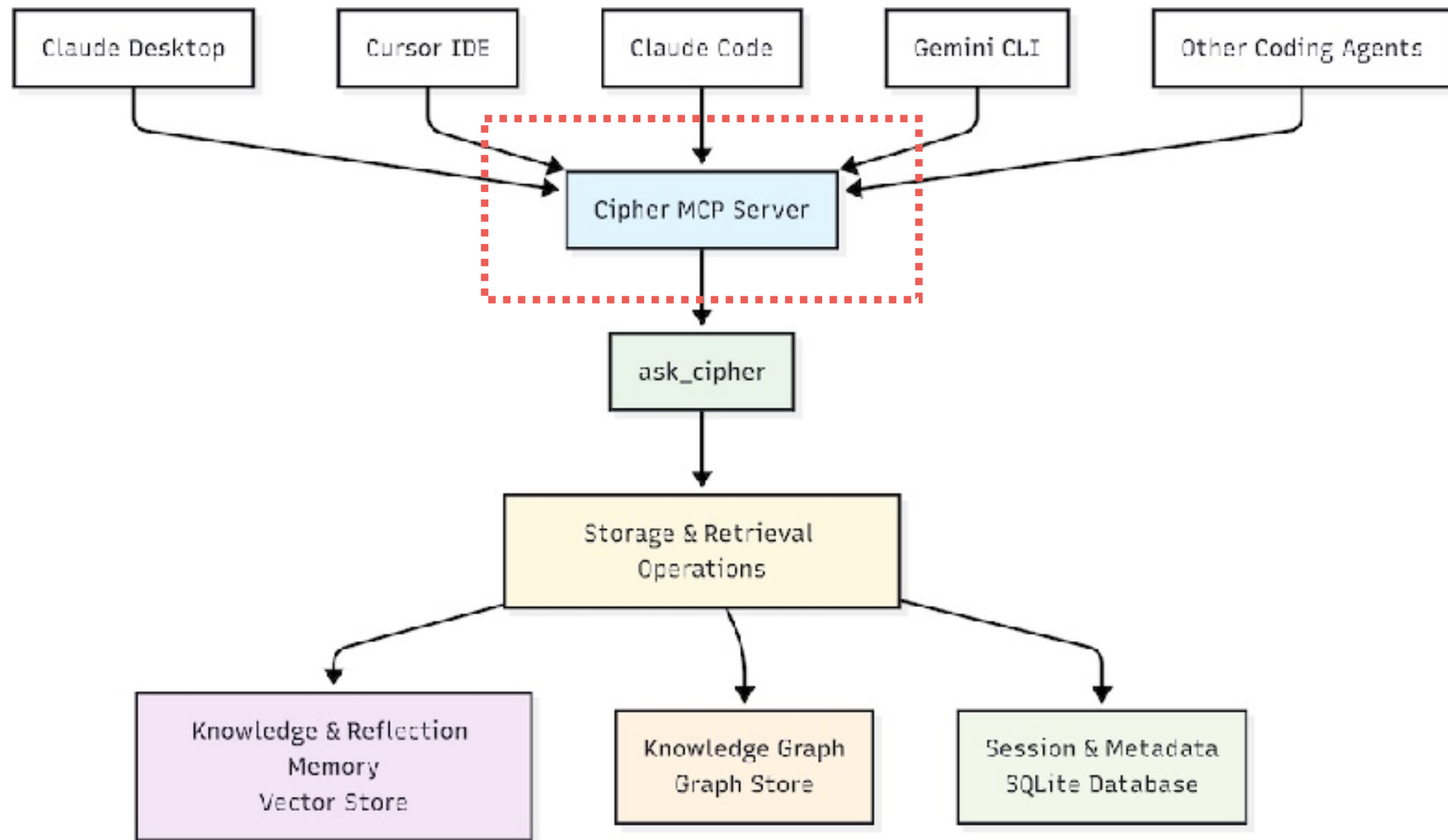
Working with file system



<https://docs.cline.bot/prompting/cline-memory-bank>



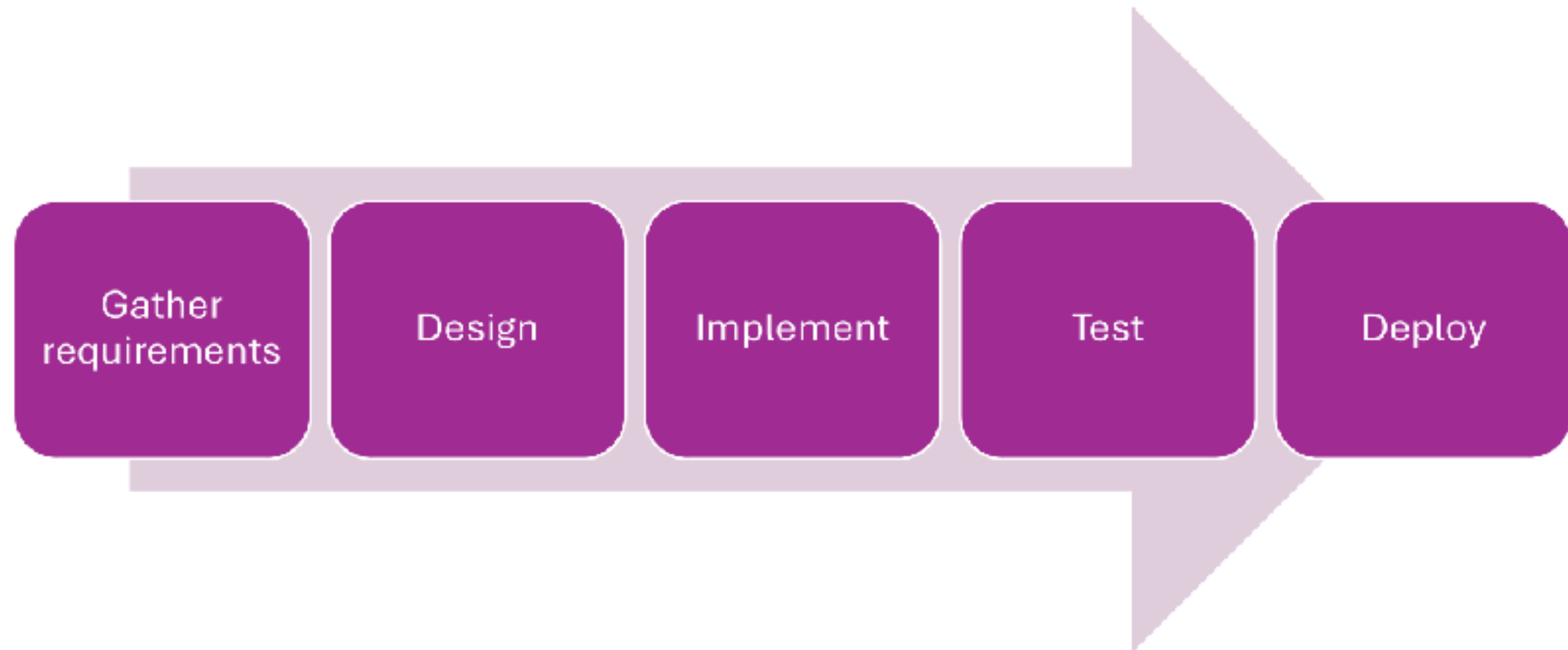
Memory Bank via MCP



<https://www.somkiat.cc/share-memory-for-ai-ide/>



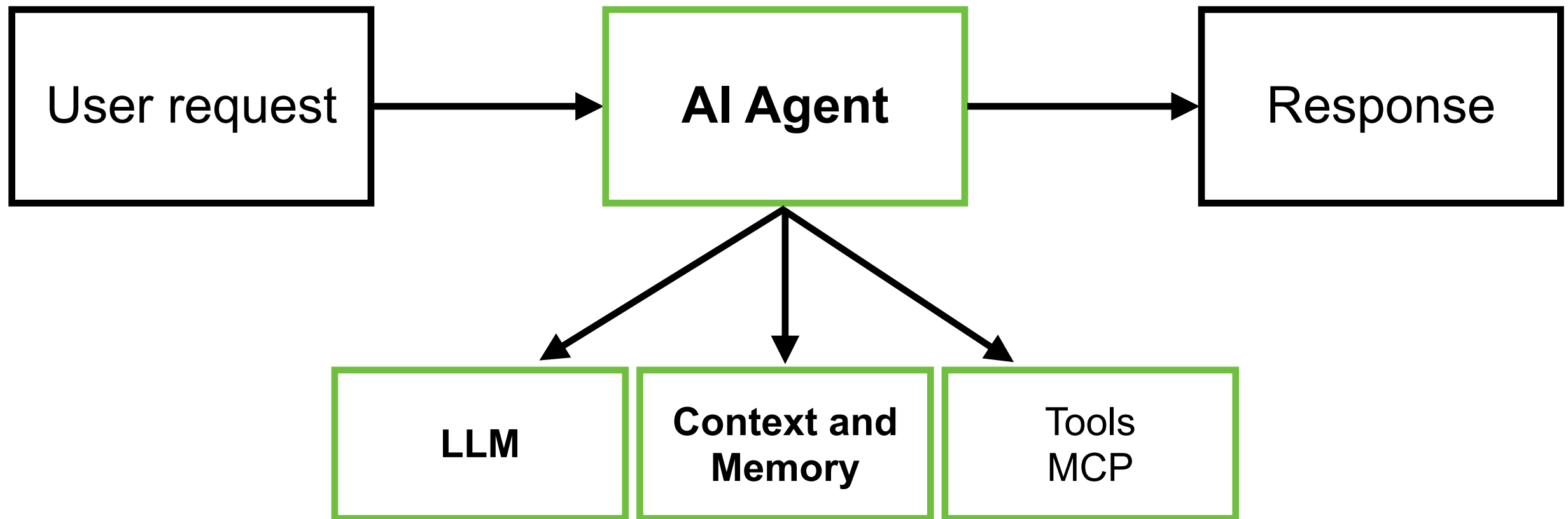
Design your AI Agents ?



<https://devblogs.microsoft.com/dotnet/introducing-microsoft-agent-framework-preview/>



Single Agent



Sub-agent and Multi-agent !! (Teamwork)

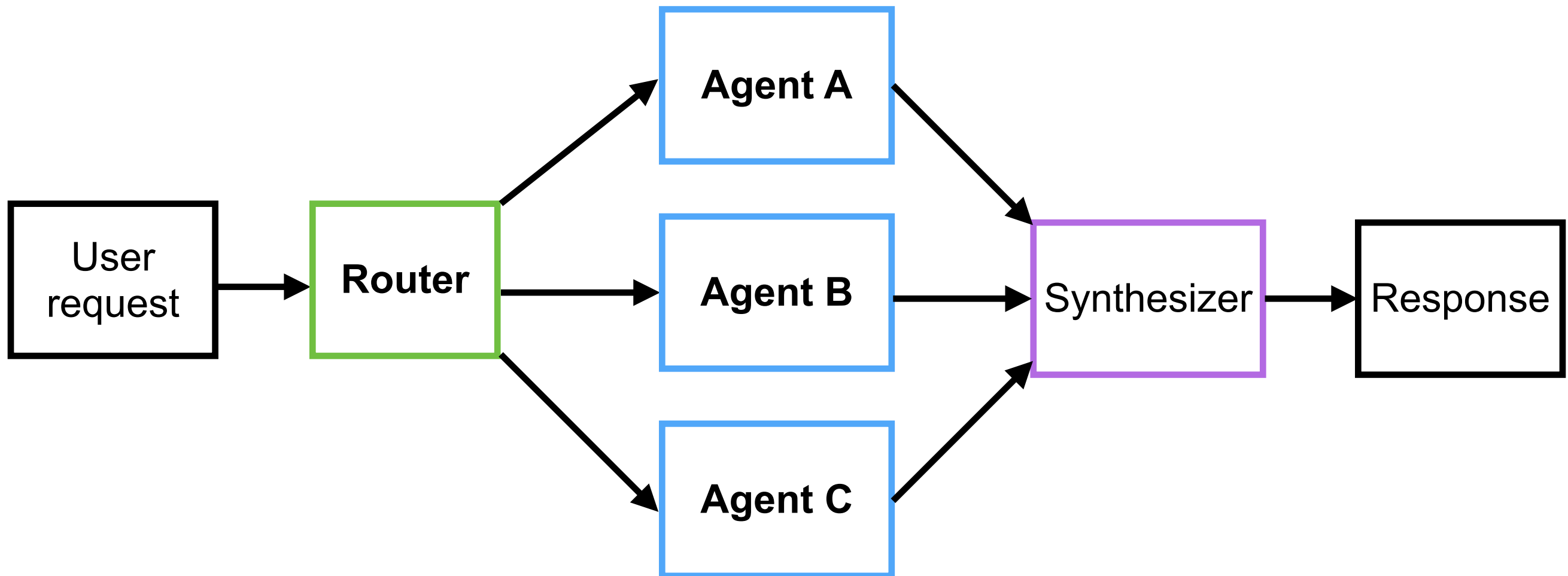
Reduce size of knowledge !!

Single Responsibility Principle



Multi AI Agents !!

Router and parallel run



Manage context !! In Software development



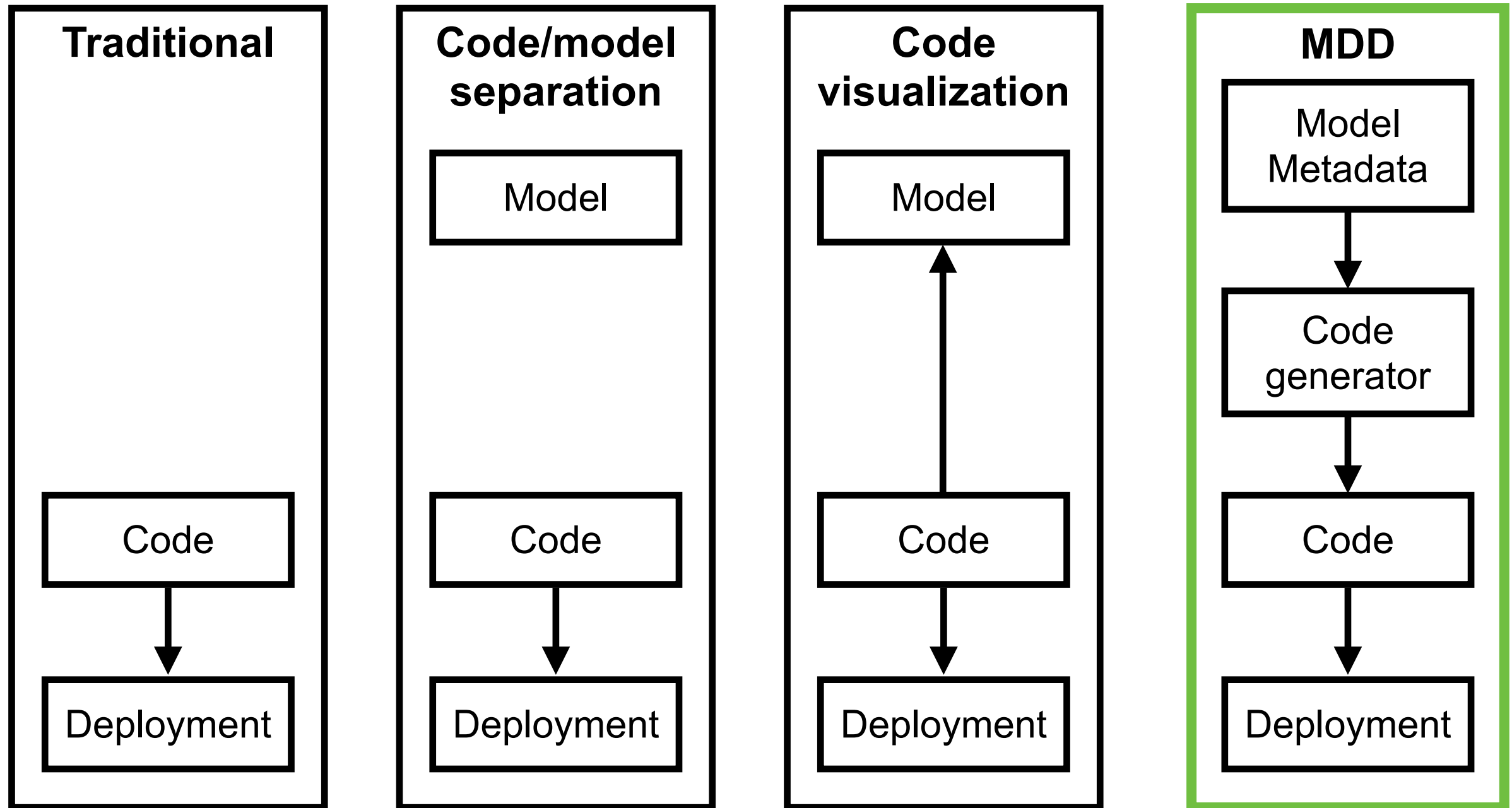
Specification ?
Architecture ?
Modeling ?
Coding convention ?
Project structure ?
Better practice ?
Technology stack ?



Model-Driven Development (MDD)



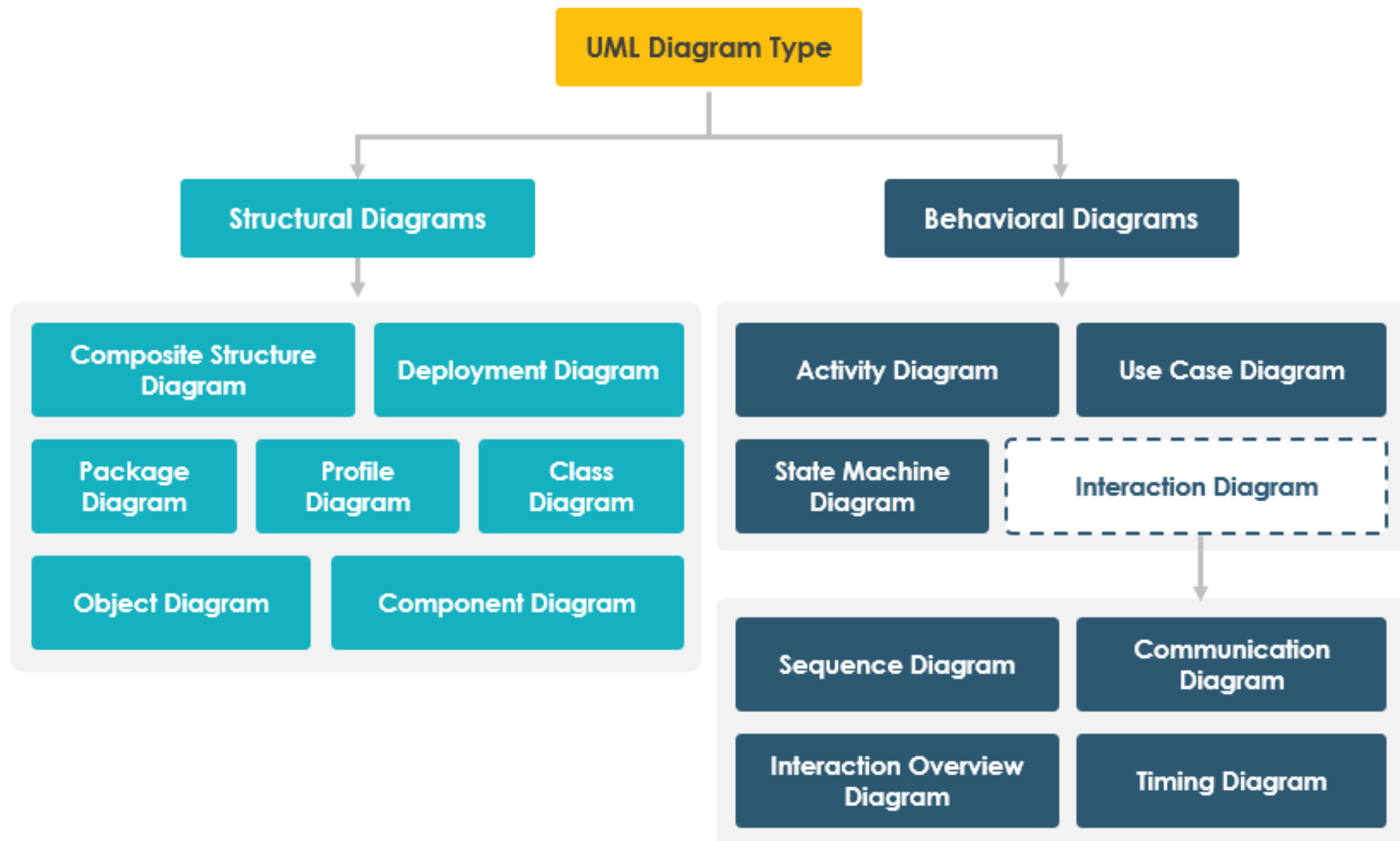
Traditional coding to MDD



<https://arxiv.org/abs/2410.18489>



UML Diagrams !!



<https://www.visual-paradigm.com/guide/uml-unified-modeling-language/behavior-vs-structural-diagram/>



Mermaid Diagram

Diagram as a Code (AI-friendly)

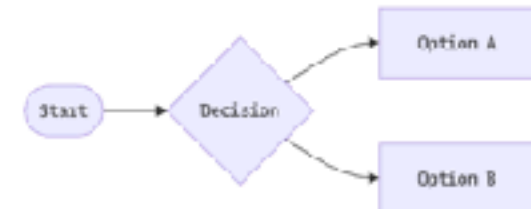
Mermaid Diagramming and charting tool

JavaScript based diagramming and charting tool that renders Markdown-inspired text definitions to create and modify diagrams dynamically.

Try Editor

Get started

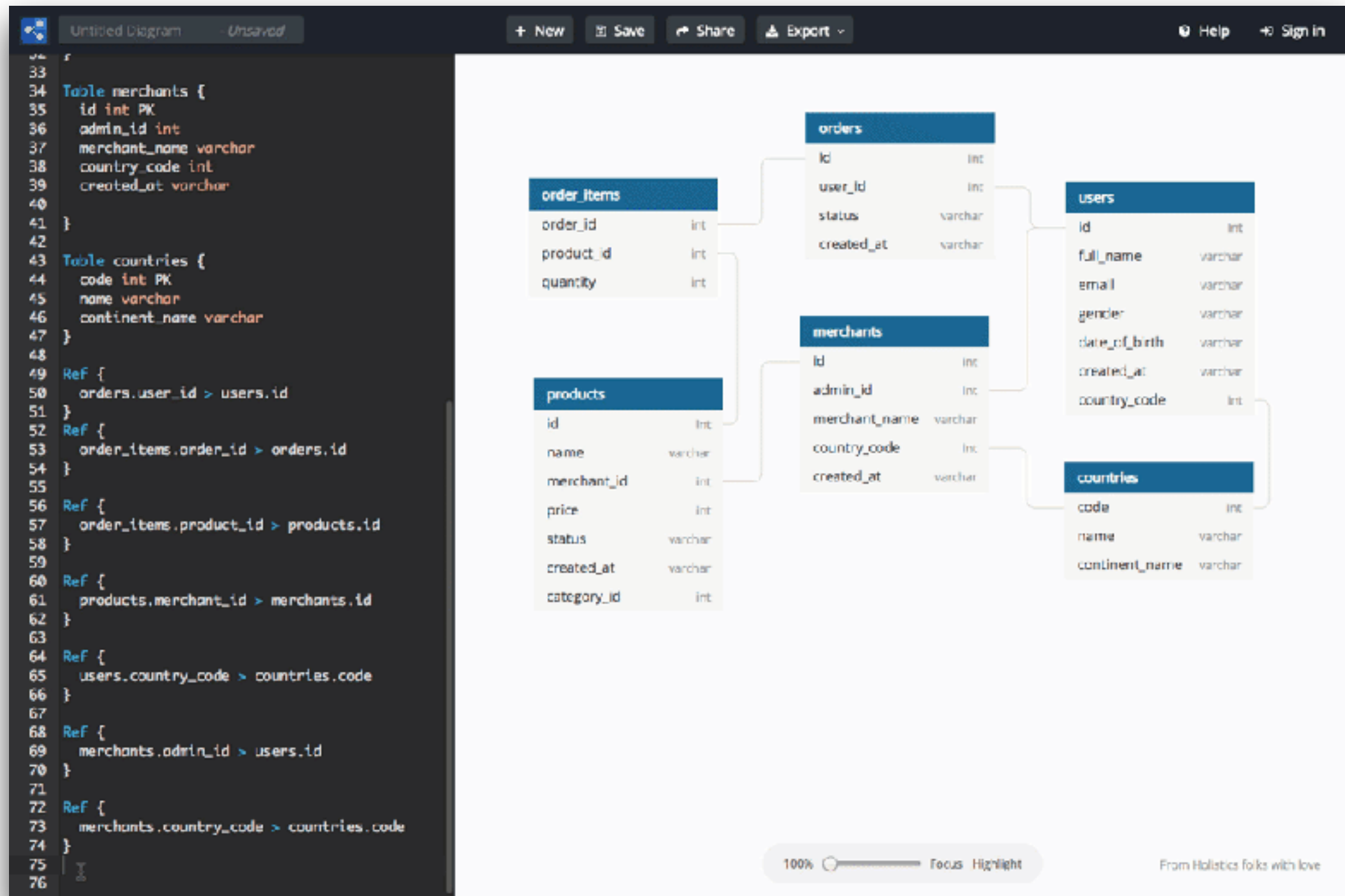
```
1 flowchart LR
2   A(["Start"])
3   A --> B["Decision"]
4   B --> C["Option A"]
5   B --> D["Option B"]
```



<https://mermaid.js.org/>



ER Diagrams for Database !!



<https://dbml.dbdiagram.io/home/>



Swagger or OpenAPI

Standardized specification for describe REST APIs

The screenshot displays the Swagger.io homepage. On the left, a hero section titled 'The Future of AI Relies on API Quality' features a green highlight on 'API Quality'. Below the title, text states: 'Swagger enables design, governance, and testing across the full AI-enabled API lifecycle, ensuring quality at every step.' and 'Build APIs ready for humans, LLMs, agents, and continuous innovation.' Two buttons are present: 'Find your tool' and 'Read the docs →'. At the bottom left, it says 'TRUSTED BY'. On the right, a preview of a 'pet' API specification is shown, listing several endpoints with their HTTP methods, paths, and descriptions.

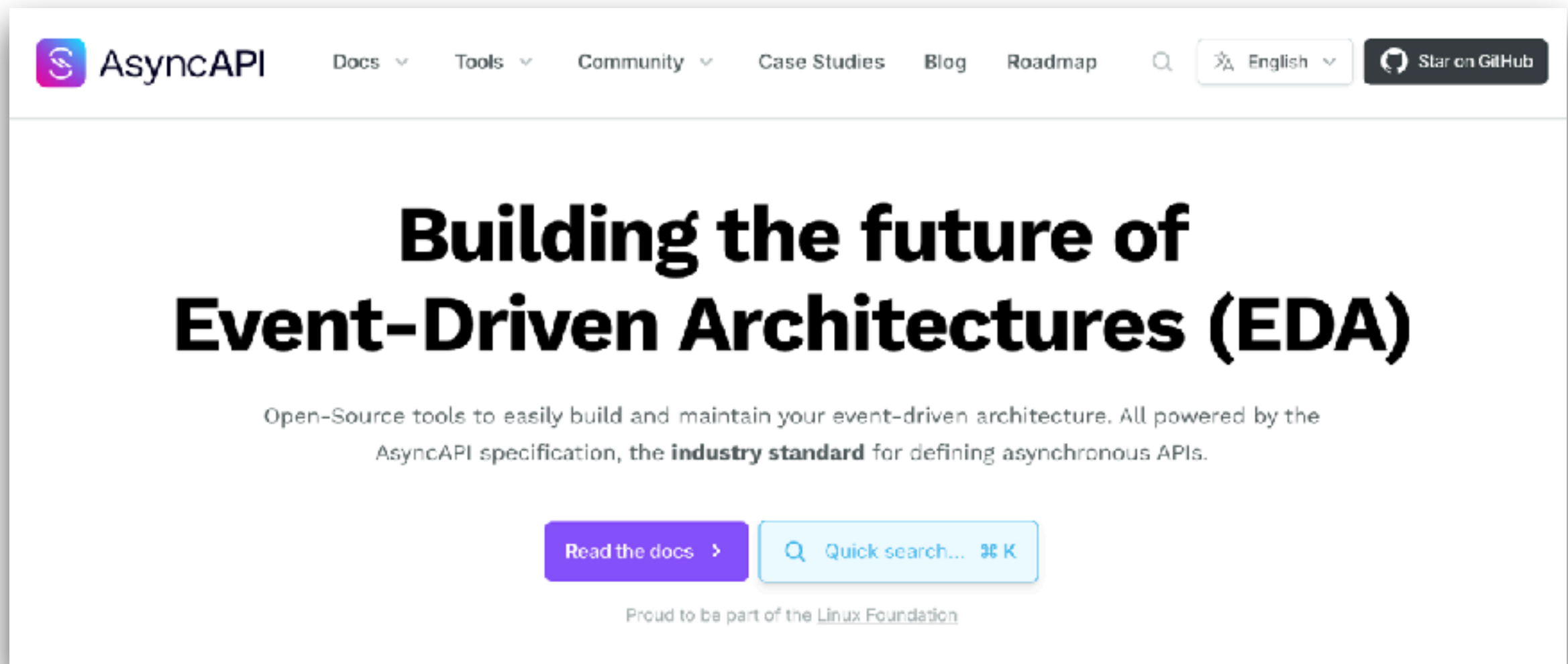
Method	Path	Description
POST	/pet/{petId}/uploadImage	uploads an image
POST	/pet	Add a new pet to the store
PUT	/pet	Update an existing pet
GET	/pet/findByStatus	Finds Pets by status
GET	/pet/{petId}	Find pet by ID
POST	/pet/{petId}	Updates a pet in the store with form data
DELETE	/pet/{petId}	Deletes a pet

<https://swagger.io/>



AsyncAPI

Standardized specification for describe asynchronous and event-driven APIs (Kafka, AMQP, WebSocket)



<https://www.asyncapi.com/>



JSONSchema

Declarative language used to annotate and validate the structure, constraints and data type of JSON documents

Data validation

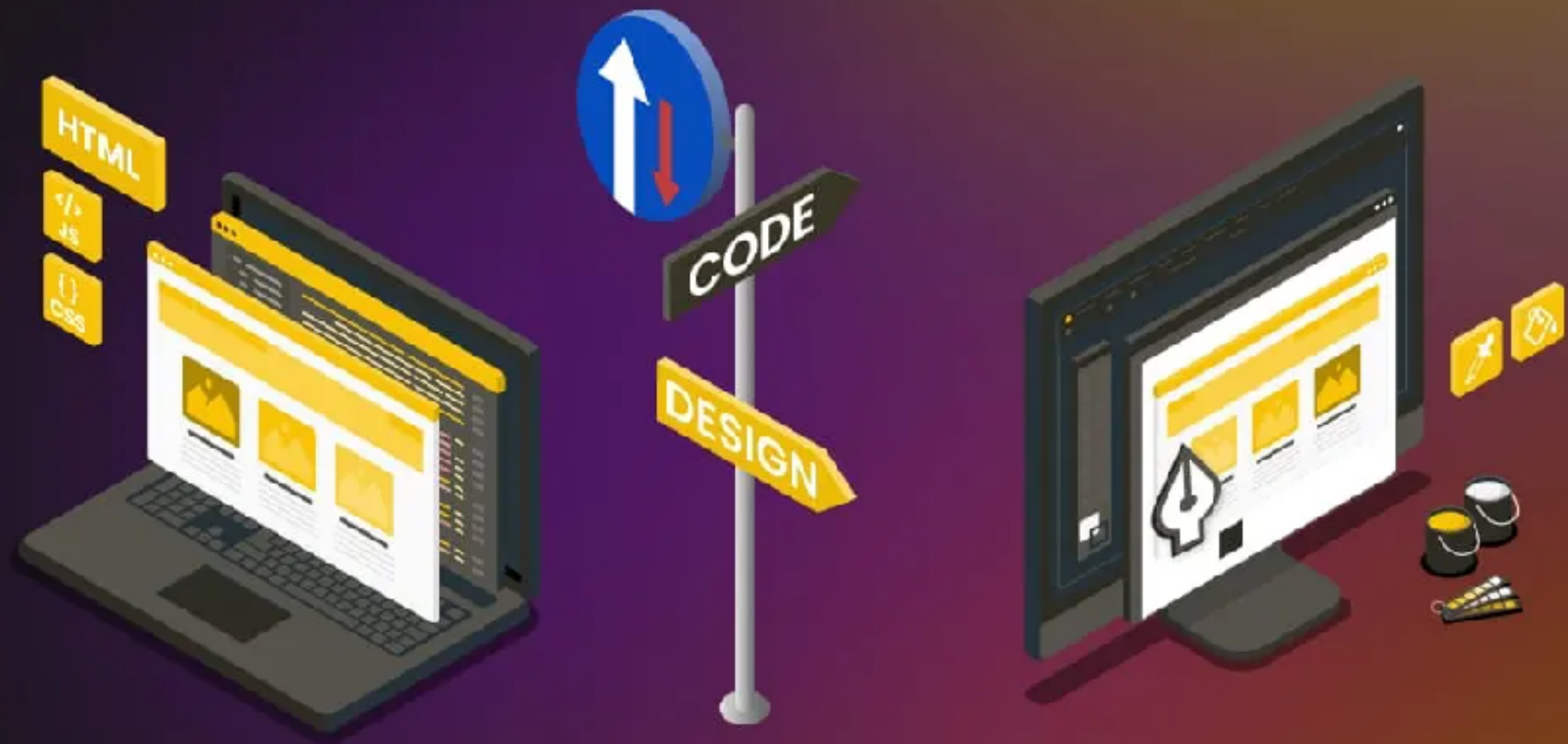
Documentation

Automated
testing

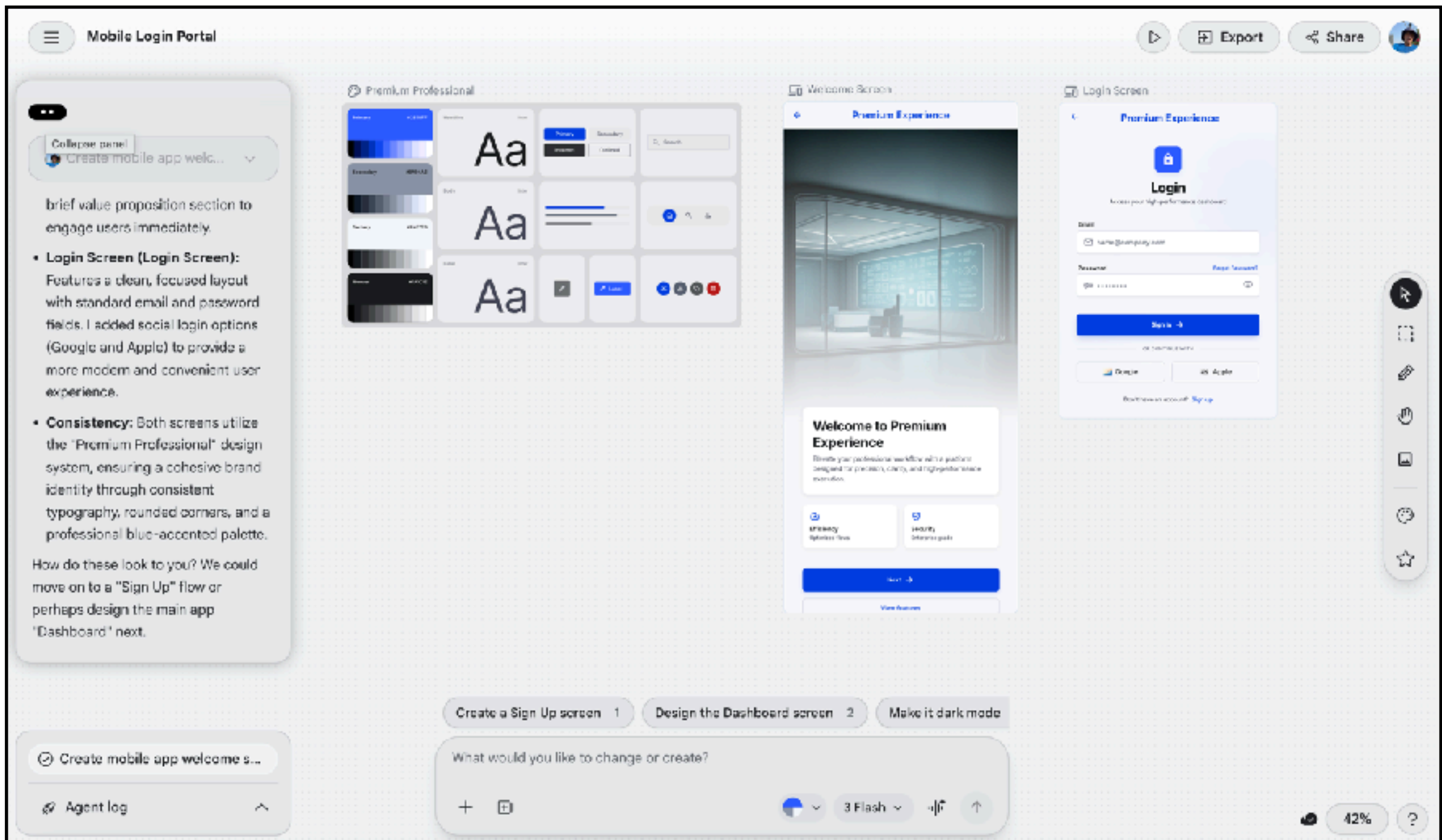
<https://json-schema.org/>



Design to Code Code to Design



Google Stitch



<https://stitch.withgoogle.com/>



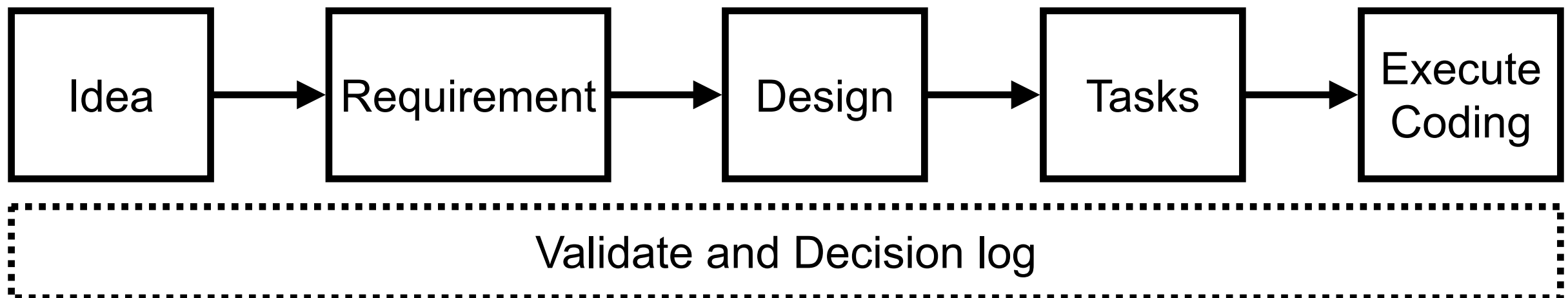
Spec-Driven Development (SDD)



SDD ?

Specification or Documentation-First

Modern software development approach where detailed, structure specifications are created before writing code

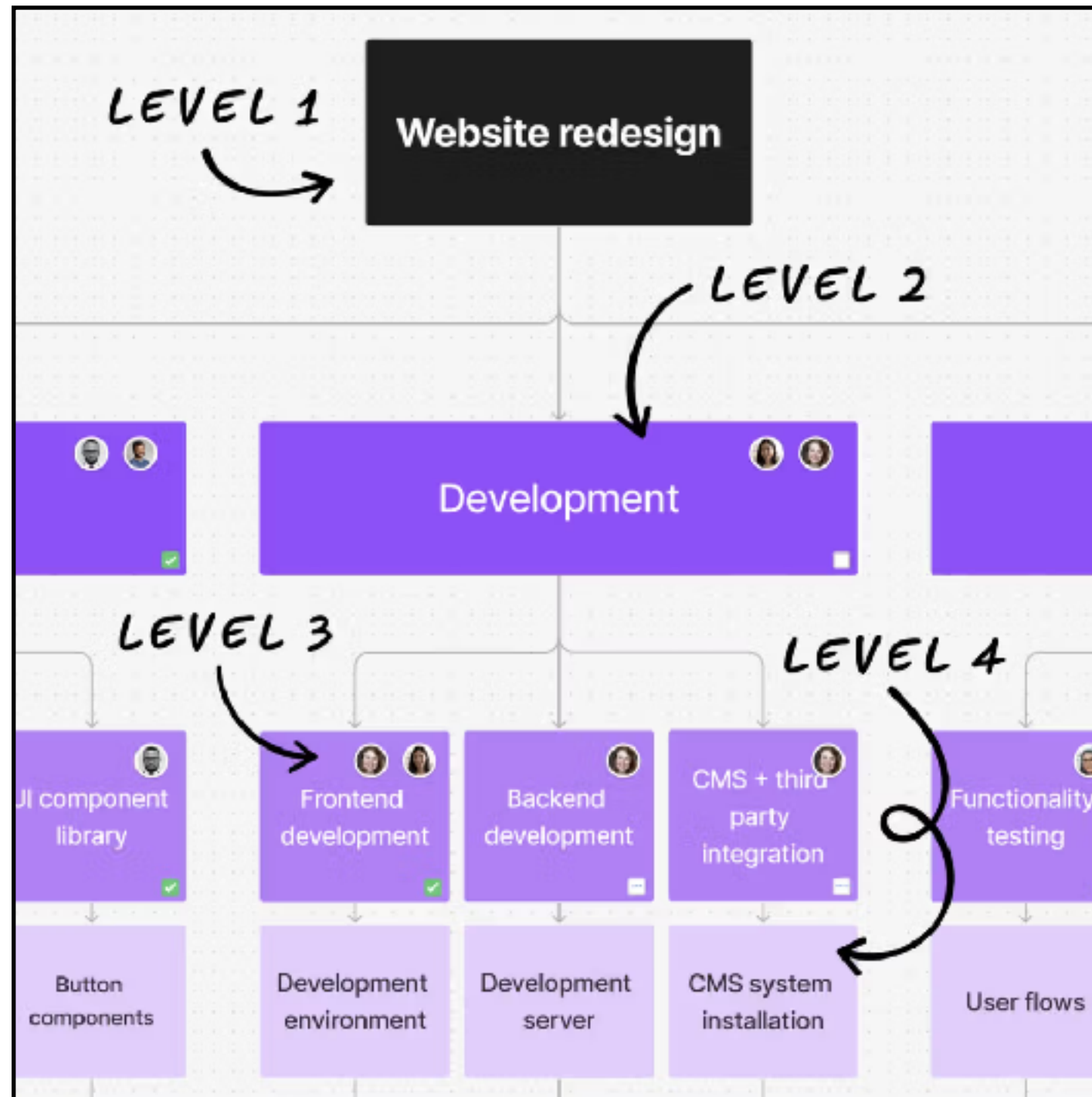


Types of Specification ?

User requirements
Design documents
Interface contracts (UI, API)
Data schemas
Validation rules
Data/business flows
Security policy and constraints
Resources and performance constraints



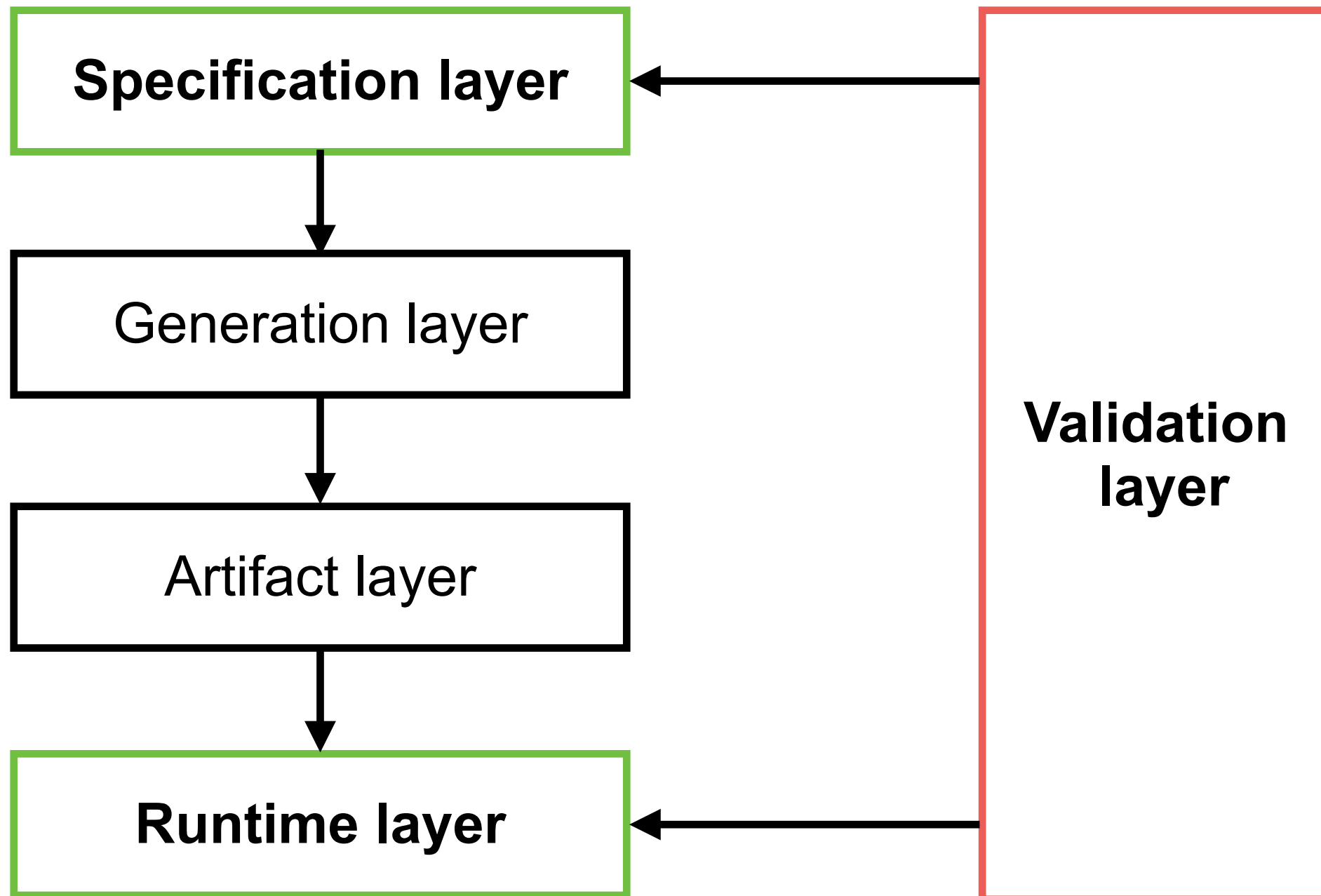
Work Break Down (WBS)



<https://www.figma.com/resource-library/work-breakdown-structure/>



SSD Architecture

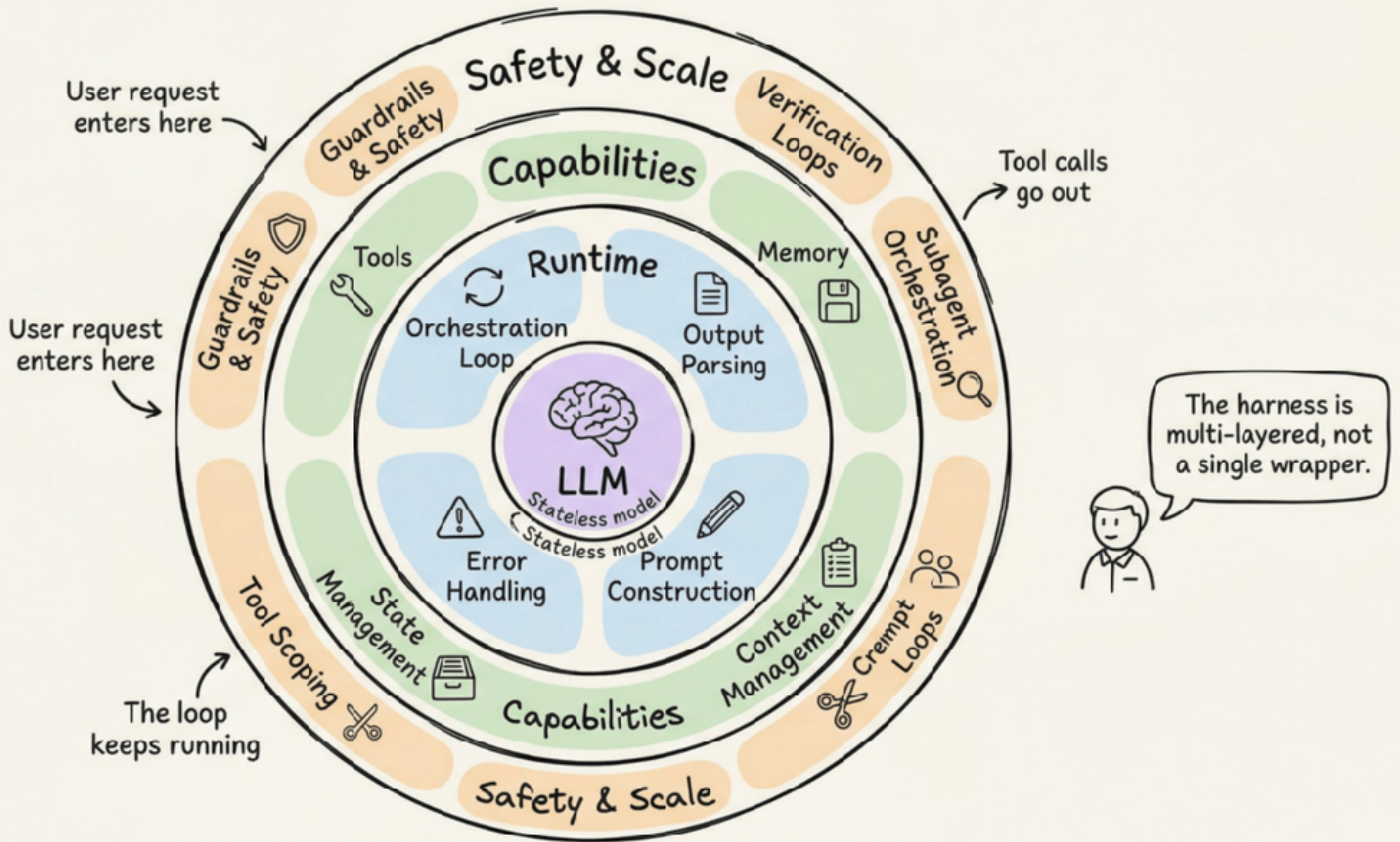


<https://www.infoq.com/articles/spec-driven-development>



MDD + SDD == Better result

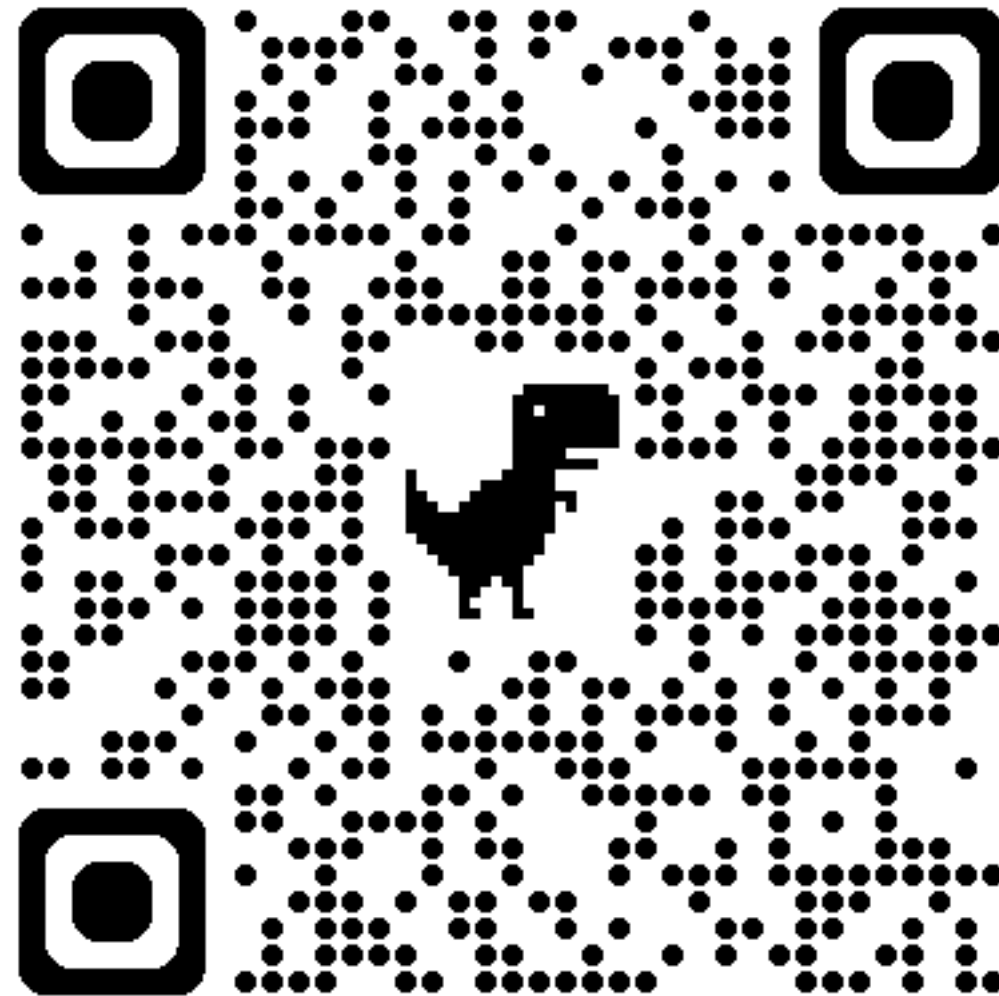




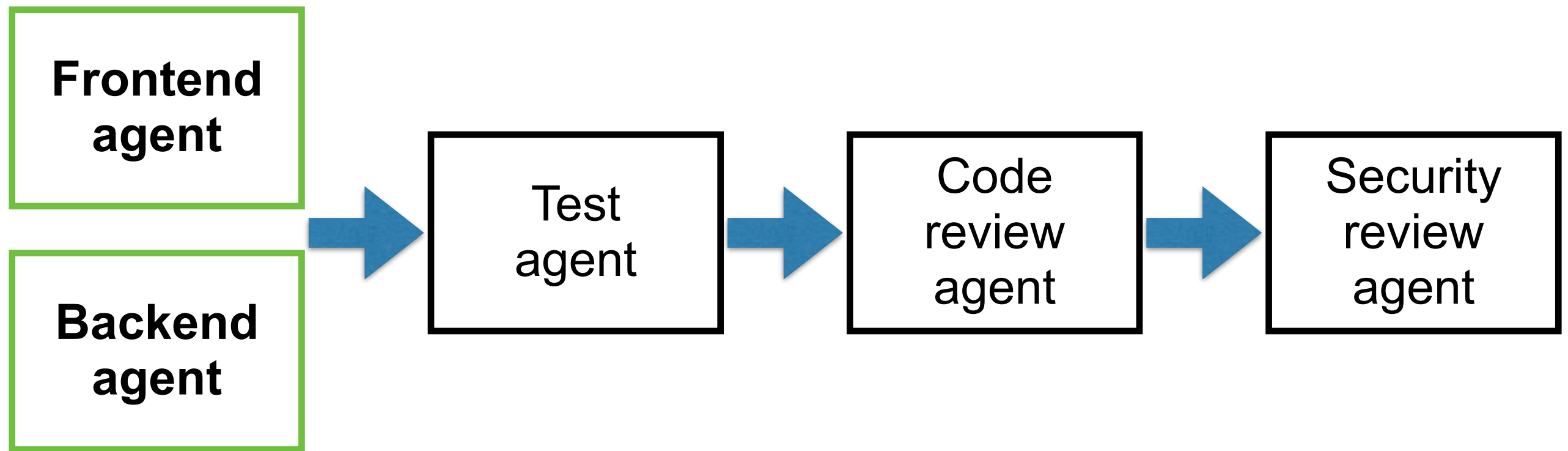
<https://blog.dailydoseofds.com/p/evolution-of-agent-landscape-from>



Demo time



Design your Agents



Manage memory and knowledge of project !!



Frontend
Mobile application

Backend
RESTful API + Database

Write specification

Coding with AI Agent

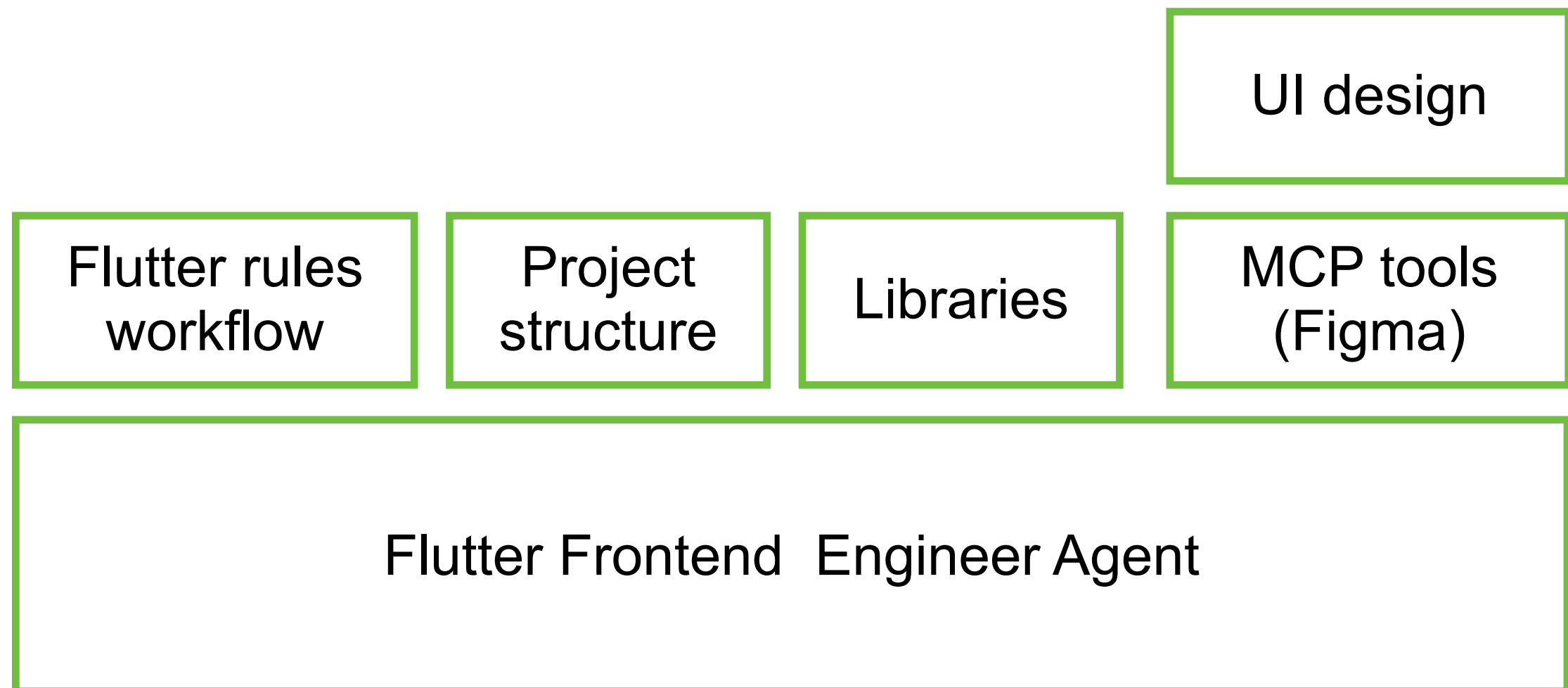
Run, Test and Verify !!

Version management system

<https://github.com/up1/demo-sharing-ai-agent-202605>

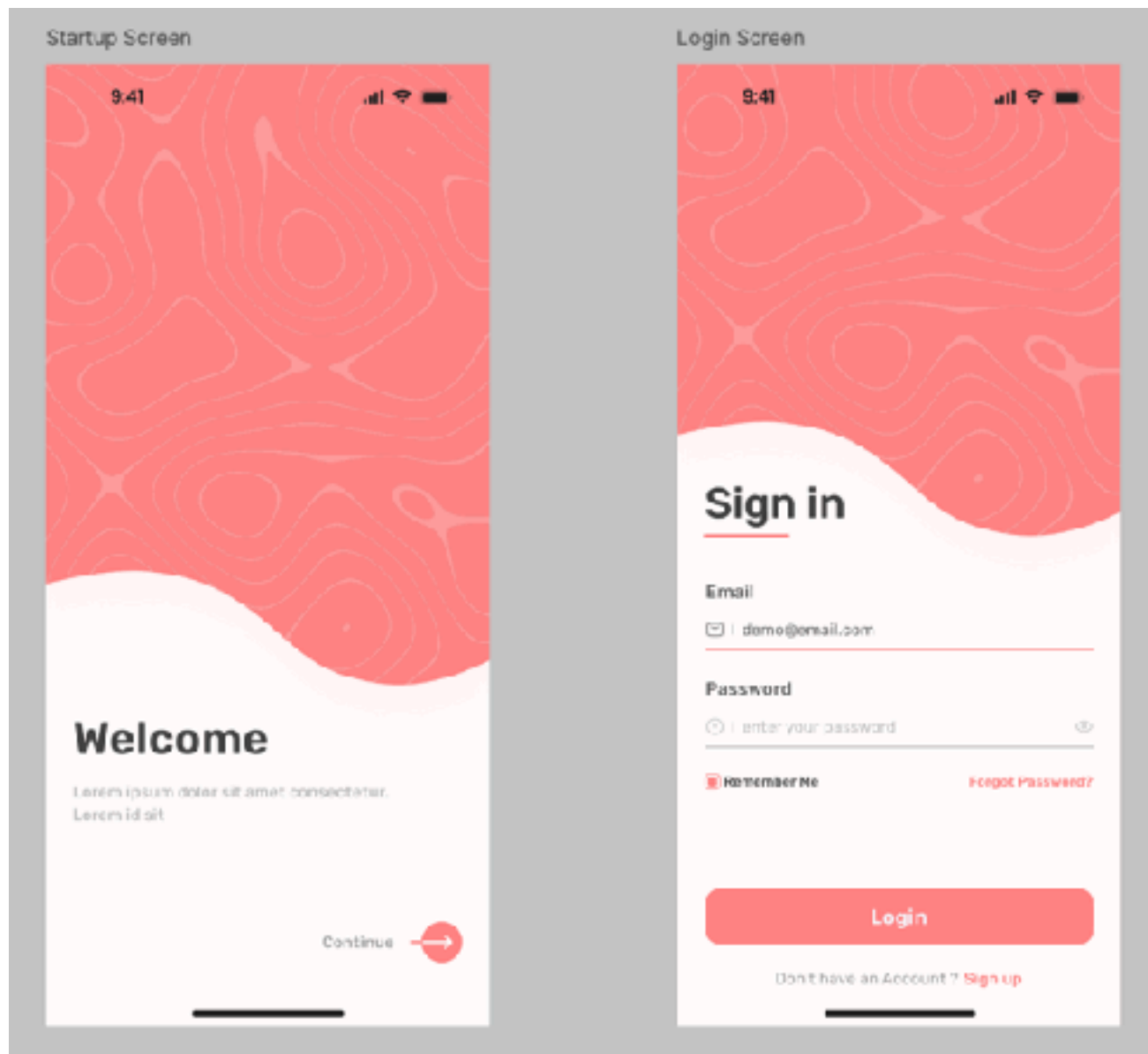


Frontend Agent (1)



User Interface Design

Design to Code technique



HTML/CSS

<https://bit.ly/3P35iNU>



Frontend Agent (2)

Write specification

UI design

Flutter rules
workflow

Project
structure

Libraries

MCP tools
(Figma)

Flutter Frontend Engineer Agent



Backend Agent

Write specification

NodeJS rules
workflow

Project structure

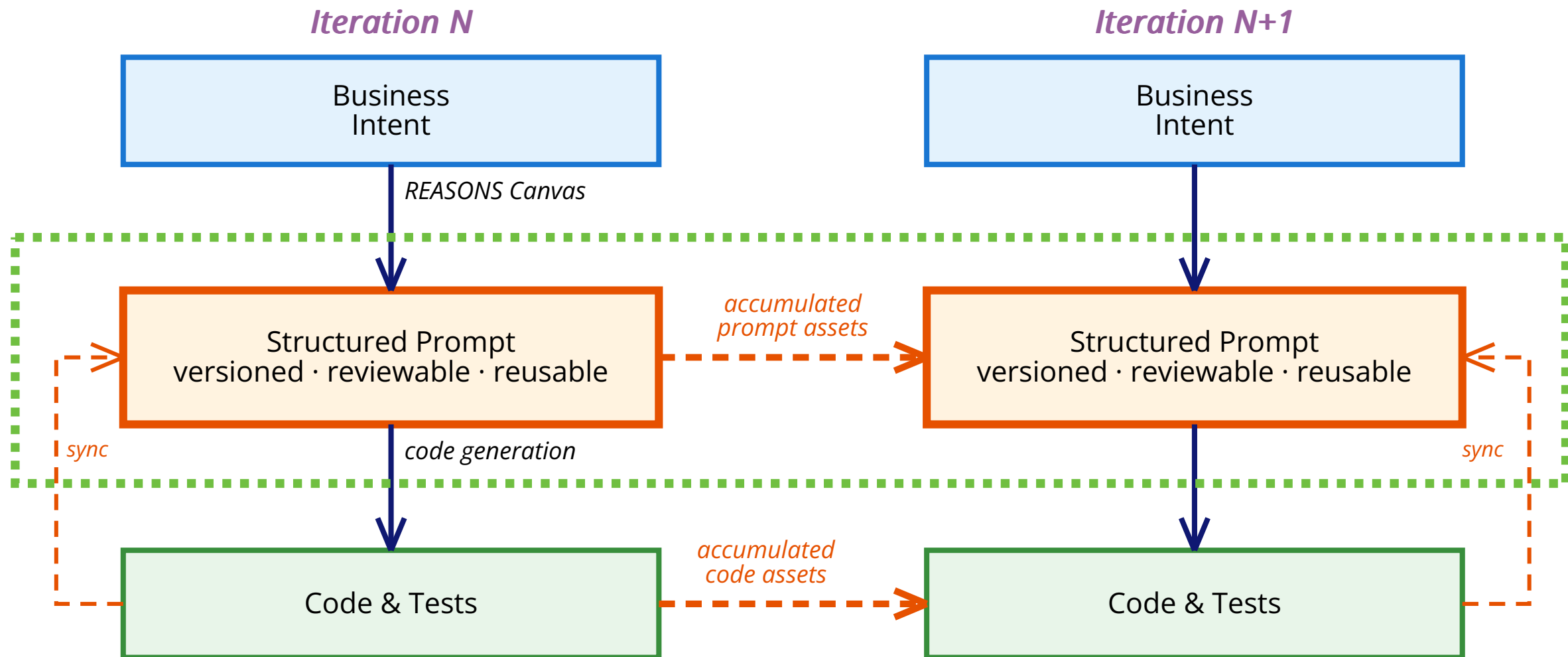
Libraries

NodeJS Backend Engineer Agent



Structure of Specification

Reliable, Repeatable and Versioning



<https://martinfowler.com/articles/structured-prompt-driven/>



Developer Survival Checklists



Developer Survival Checklists

1. Focus on logic (good or bad)
2. Audit AI Output (functional and non-functional)
3. Continuous Learning

**Focus on foundation knowledge
to effectively work with and guide AI system**



AI draft

Human craft



Slide

